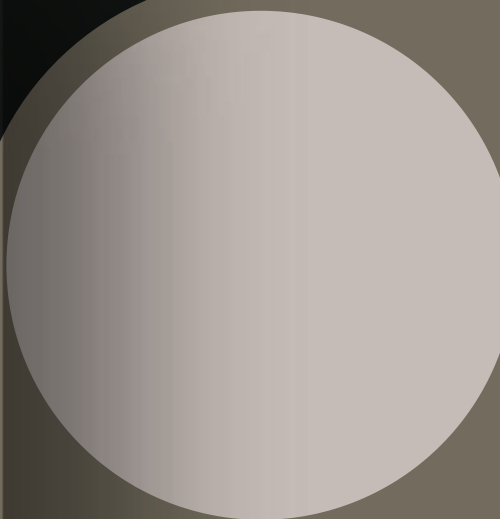




PROFESSIONAL INTERNSHIP PRESENTATION

July 25, 2026



AGENDA: MODULES 1–3 OVERVIEW

1

Module 1

Introduction to IT and Business Systems: covers IT fundamentals, business systems (ERP, CRM), IT-business strategy alignment, and cloud/ITSM basics.

2

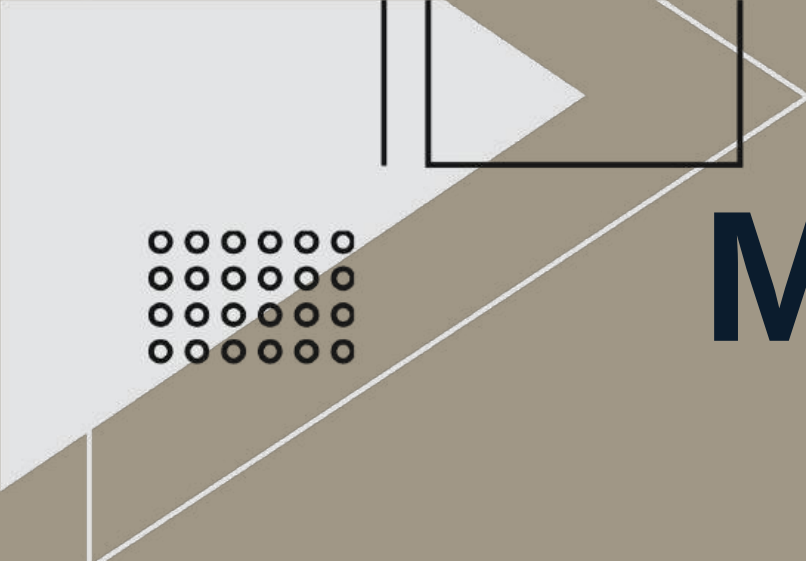
Module 2

Data Types, Sources, and Lifecycle: explores structured/unstructured data, data sources, collection methods, lifecycle stages, quality, big data, warehousing vs. lakes, and MDM.

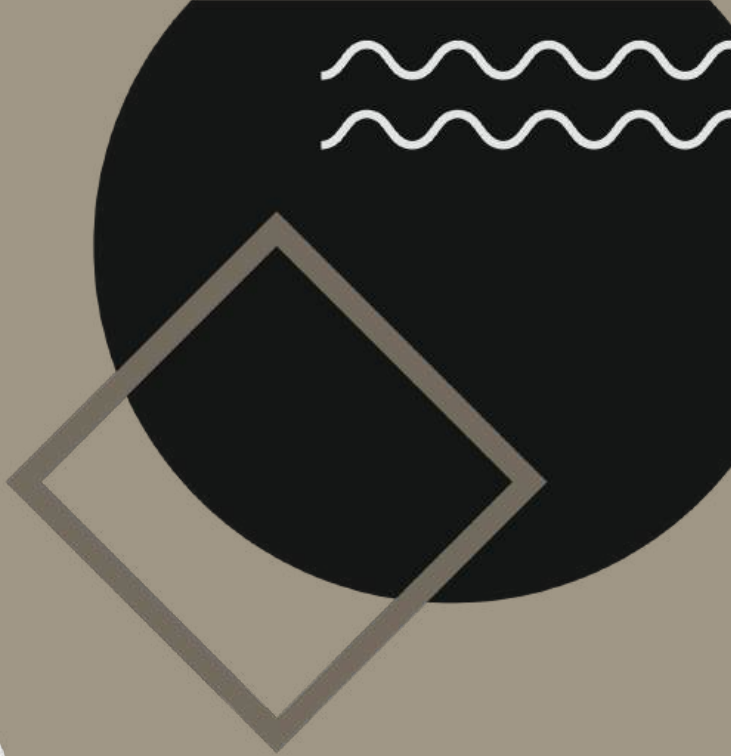
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Module 3

Introduction to Business Analyst Role: defines BA responsibilities, key skills, Agile vs. Waterfall, stakeholder management, elicitation techniques, process modeling, use cases, and gap analysis.



MODULES 4-6 OVERVIEW



1

Data vs. Information

Explore how raw data is transformed into meaningful information and knowledge, and understand the value of information in supporting evidence-based decisions.

2

Data-Driven Decision Making

Learn how organizations leverage data analytics, key metrics, and visualization to drive strategic decisions, optimize performance, and improve outcomes.

3

Case Discussions

Apply concepts through real-world examples, case studies, and practical scenarios that illustrate successful data and IT strategies in action.



MODULE 7: DATA GOVERNANCE & ETHICS

DATA GOVERNANCE



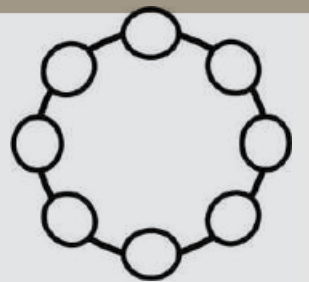
Policies, standards, and roles that ensure data quality, security, and compliance across the organization.

DATA ETHICS



Responsible use of data: privacy, consent, fairness, transparency, and minimizing bias in analytics.

PRESENTATION FLOW



7 modules from IT systems and data management to business analysis, decision making, and governance.

MODULE 1

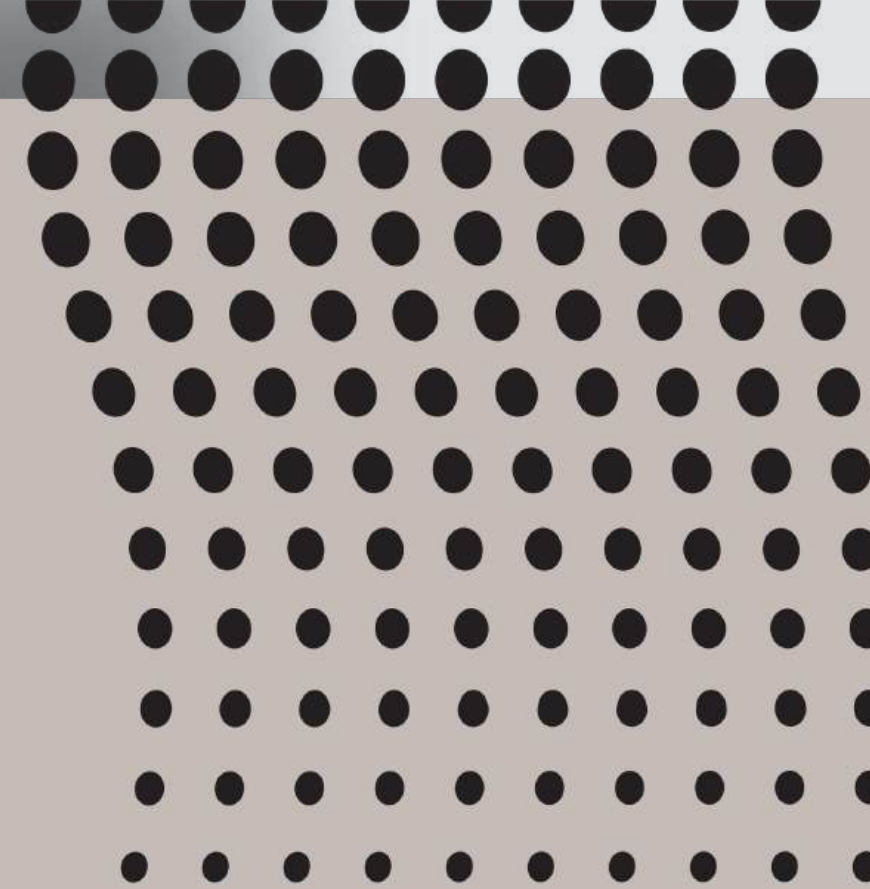
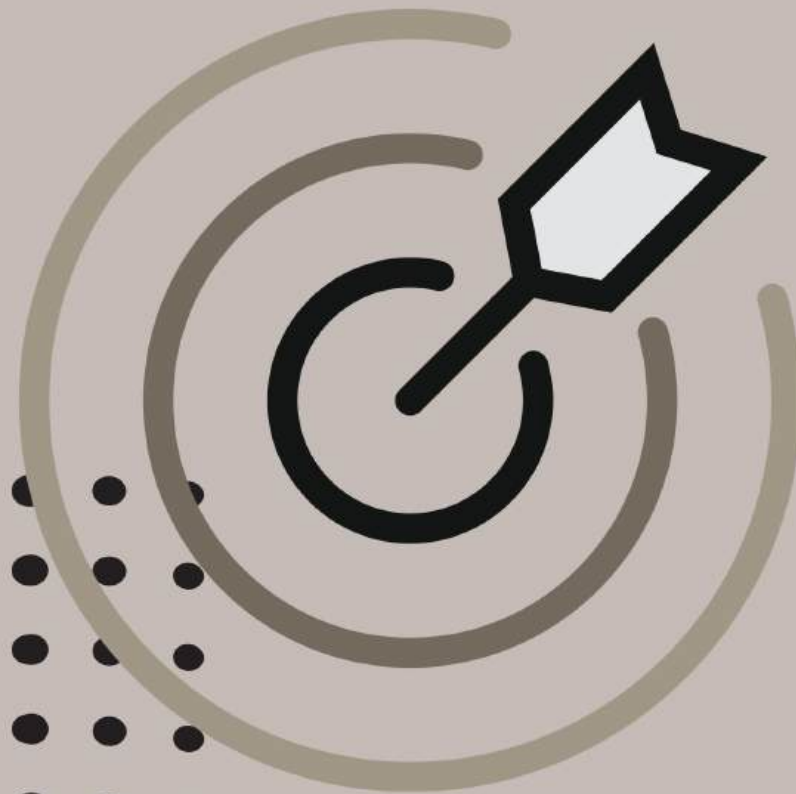
Introduction to IT and Business Systems

Foundation concepts for understanding how information technology and business systems work together to support organizational goals.

IT & BUSINESS SYSTEMS

FUNDAMENTALS OF INFORMATION TECHNOLOGY

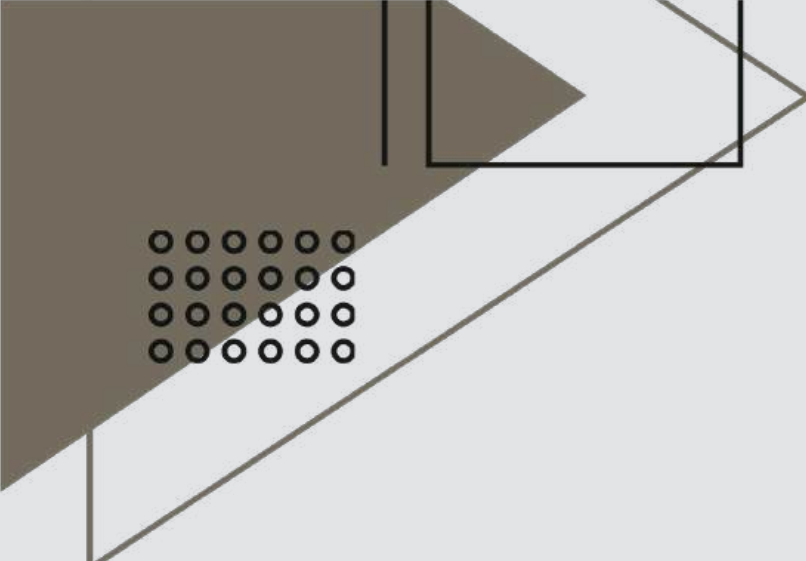
Information Technology (IT) is the use of computer systems, hardware, software, and networks to manage, process, and exchange information. It enables businesses to operate efficiently, communicate seamlessly, and make informed decisions in a modern digital environment.



Key IT Components:

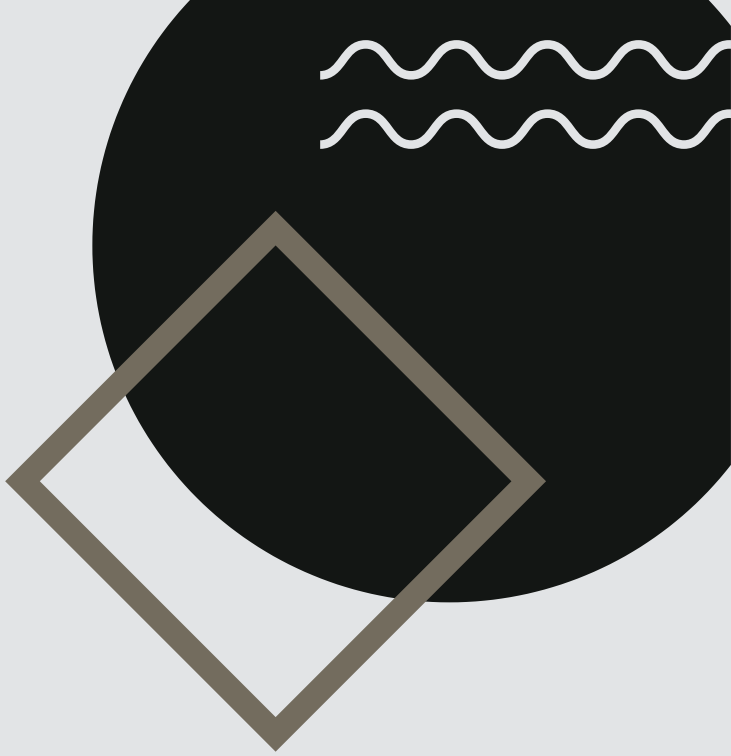
- Hardware: physical devices such as servers, computers, and mobile devices
- Software: applications and operating systems that run business processes
- Networks: infrastructure that connects systems and enables data sharing
- Role in Business: IT supports operations, drives collaboration, and enables strategic innovation across the organization.

IT IN MODERN BUSINESS



IT ROLES IN BUSINESS OPERATIONS

Transforms data into actionable insights that improve forecasting, strategic planning, and operational performance. IT also enhances customer experience through personalized digital services and drives innovation by enabling new products, business models, and technologies.



Enabling Communication & Collaboration

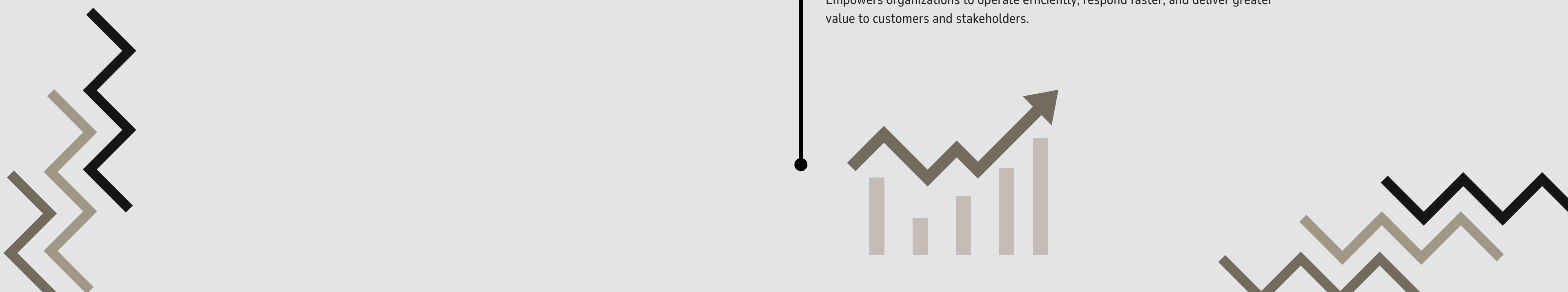
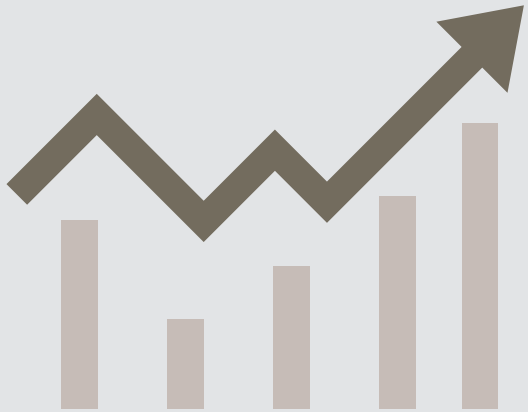
Connects teams across locations through email, video conferencing, instant messaging, and shared platforms.

Automating Tasks & Processes

Reduces manual effort, minimizes errors, and speeds up repetitive workflows using software and robotic process automation.

Supporting Decision Making

Empowers organizations to operate efficiently, respond faster, and deliver greater value to customers and stakeholders.



ERP VS CRM

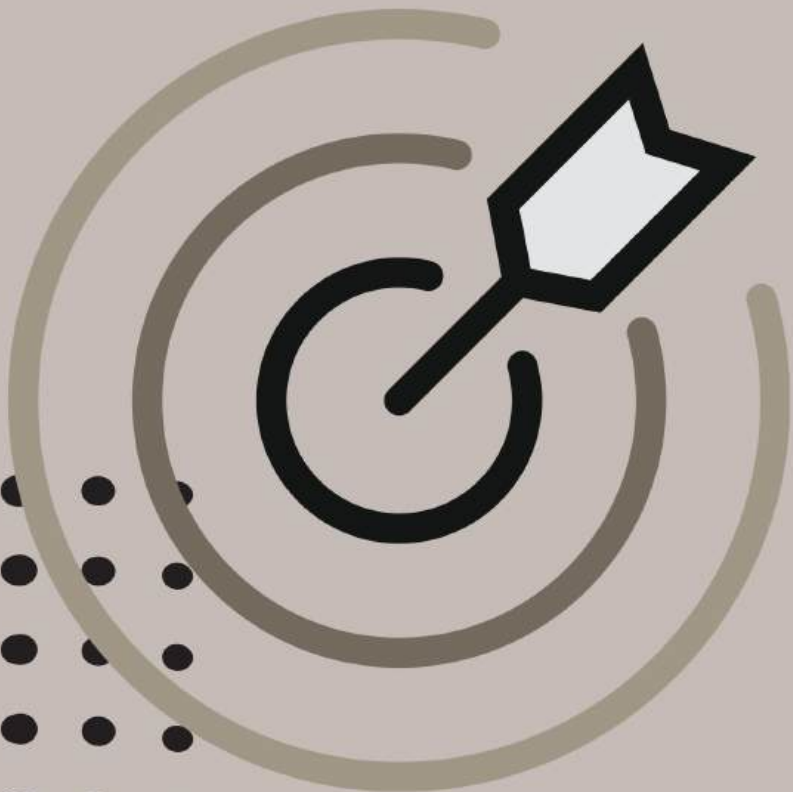
Enterprise Resource Planning (ERP) integrates internal business processes such as finance, supply chain, HR, and operations into a single unified system. It streamlines workflows, improves efficiency, and provides a centralized source of business data.

Customer Relationship Management (CRM) focuses on managing customer relationships, sales pipelines, and marketing activities. It helps organizations track leads, improve customer service, and drive revenue growth.

SYSTEMS COMPARED

IT & BUSINESS STRATEGY ALIGNMENT

Aligning IT with business goals ensures technology investments support strategic objectives such as digital transformation, cloud adoption, and operational excellence. The IT-Business Strategic Alignment Model helps organizations coordinate technology and business strategies for shared value and competitive advantage.



Strategic alignment of IT and business goals
Digital transformation and cloud adoption
initiatives
IT-Business Strategic Alignment Model overview

STRATEGIC ALIGNMENT



CLOUD MODELS & ITSM

Cloud computing transform how organizations access technology. IaaS, PaaS, and SaaS provide flexible, scalable service models, while ITSM ensures reliable delivery and cybersecurity safeguards every layer.



Cloud Computing Models

IaaS: Infrastructure as a Service—virtualized servers, storage, networking

PaaS: Platform as a Service — development platforms and middleware

SaaS: Software as a Service — ready-to-use applications over the internet

IT Service Management

ITSM aligns IT services with business needs through structured processes

Includes incident management, change management, and service delivery

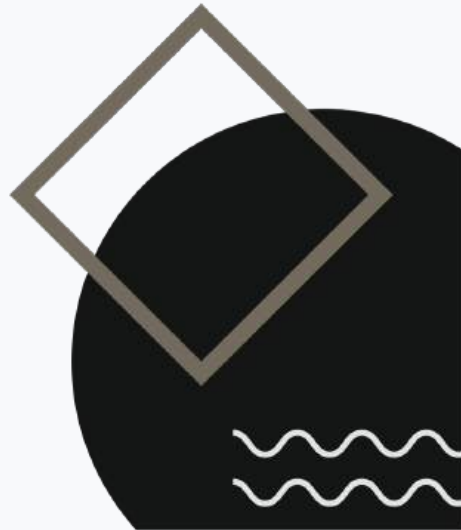
Supports continuous improvement and reliable service operations

Cybersecurity Essentials

CIATriad: Confidentiality, Integrity, Availability

Protects data, systems, and services against threats and breaches

Essential foundation for secure cloud and IT operations

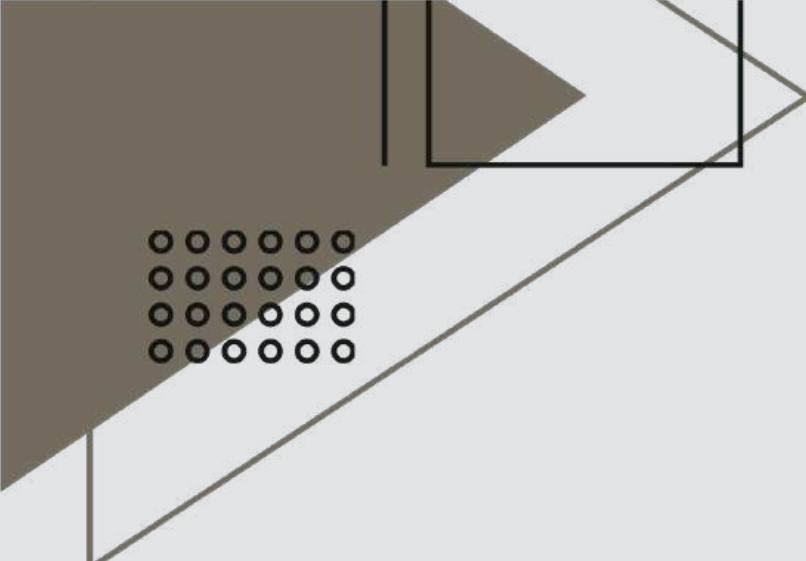


MODULE 2

Data Types, Sources, and Lifecycle

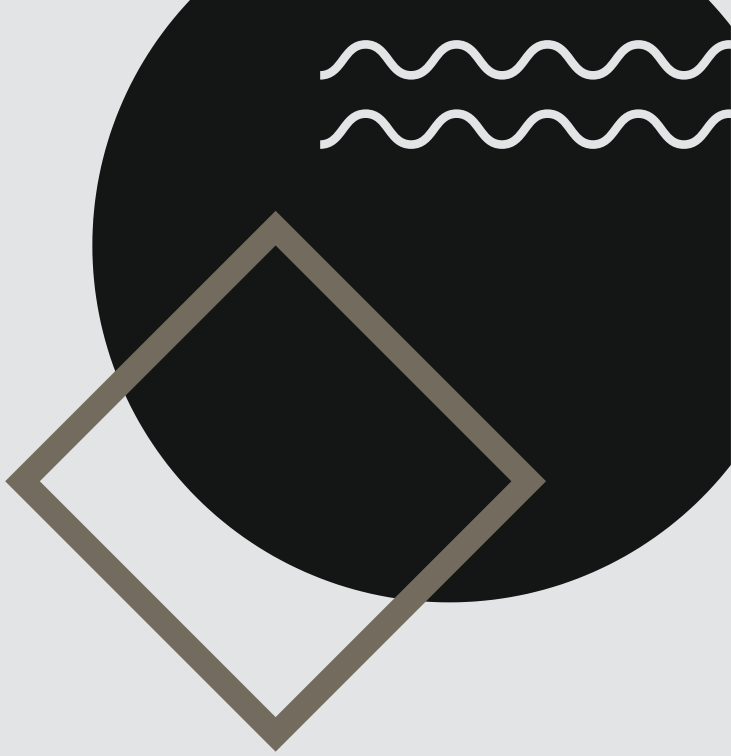
Understanding data fundamentals: how data is classified, where it comes from, and how it is managed across its lifecycle.

DATA FUNDAMENTALS



UNDERSTANDING DATA TYPES

Recognizing the differences between structured, unstructured, and semi-structured data is essential for effective data management, storage, and analysis across modern business systems.



Structured Data

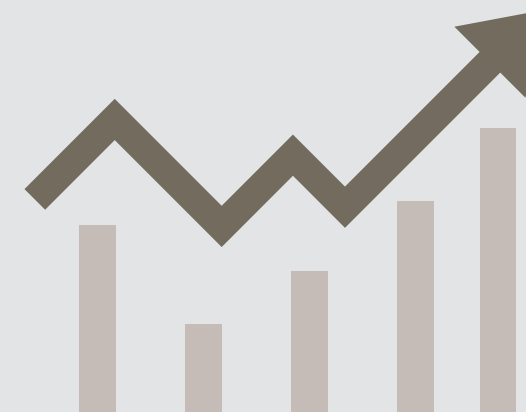
Highly organized data stored in fixed formats, such as relational databases and spreadsheets, making it easy to query and analyze.

Unstructured Data

Information without a predefined data model, including emails, videos, images, and social media content, requiring advanced processing to analyze.

Semi-Structured Data

Data that does not conform to a rigid schema but includes tags or metadata, such as JSON and XML files, offering flexibility between structured and unstructured formats.



COMMON DATA SOURCES

INTERNAL SOURCES



Internal sources such as ERP and CRM systems, along with operational logs, capture structured records of transactions, customer interactions, and system activity.

EXTERNAL SOURCES



External sources include market data feeds, social media platforms, and public datasets that provide broader context and competitive insight.

ACCESS METHODS



Databases and APIs serve as the primary access methods, enabling secure, reliable retrieval and integration of data across systems.



DATA COLLECTION METHODS

Explore the key methods used to gather data for business systems and analytics, including manual entry, automated capture, web scraping, APIs, and surveys.



Manual Entry & Automated Capture

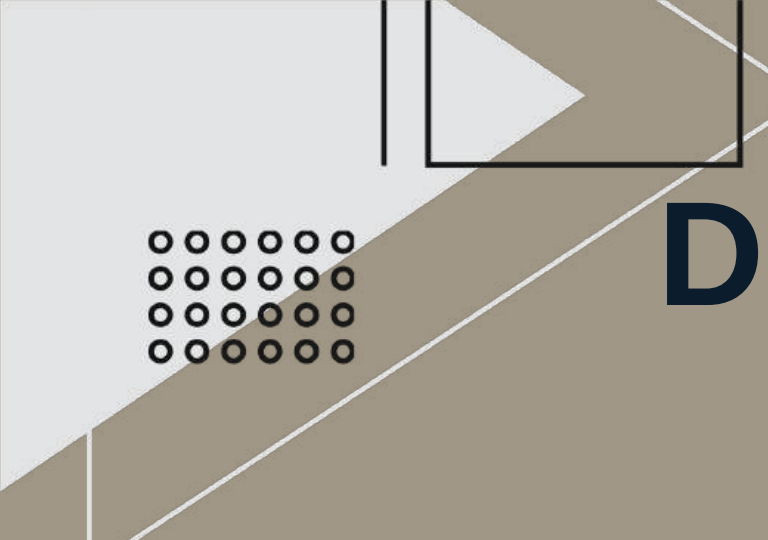
- Manual entry: Data entered by users into forms, databases, or spreadsheets.
- Automated capture: IoT devices and sensors collect real-time data with minimal human intervention.

Web Scraping & APIs

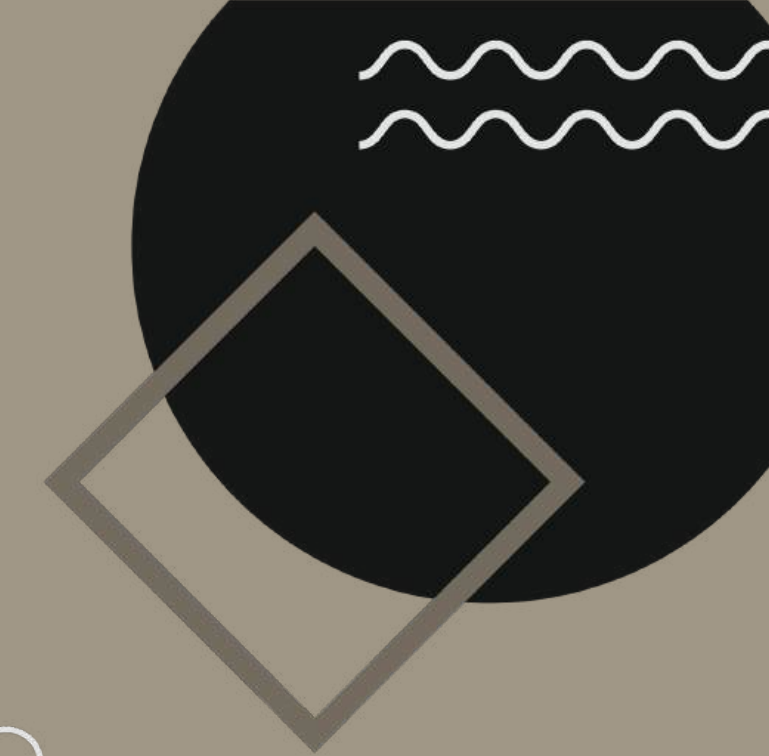
- Web scraping: Automated extraction of data from websites for analysis.
- APIs: Standardized interfaces that allow systems to request and exchange data securely.

Surveys & Best Practices

- Surveys and feedback forms capture structured opinions from users and stakeholders.
- Best practices: Ensure data quality, privacy, compliance, and validation at every collection stage.



DATA LIFECYCLE MANAGEMENT



1

Creation

Data is generated from internal systems, external feeds, IoT sensors, and user interactions. Quality starts at creation with accurate capture and clear metadata.

2

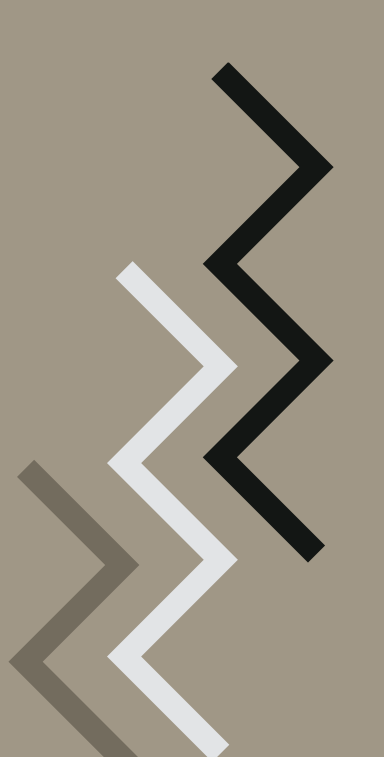
Usage

Data is analyzed, shared, and applied to business decisions. Proper usage ensures security, compliance, and value creation across the organization.

3

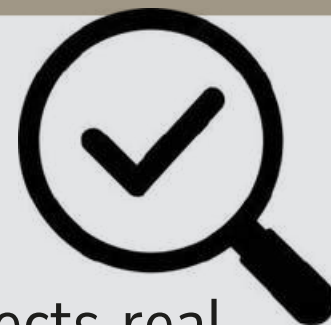
Deletion

Outdated data is archived or securely deleted according to retention policies. This reduces risk, ensures compliance, and optimizes storage costs.



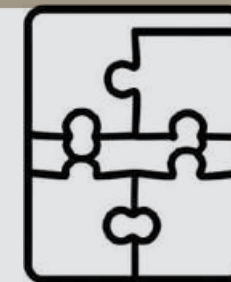
DATA QUALITY AND INTEGRITY

ACCURACY



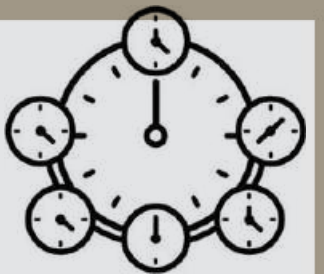
Data correctly reflects real-world values and facts, free from errors or bias.

COMPLETENESS

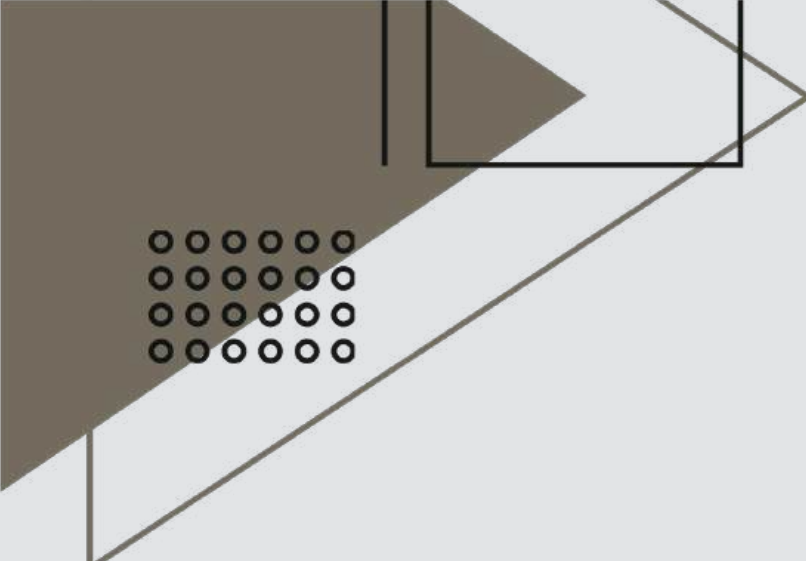


All required data fields and records are present without missing or null values.

CONSISTENCY & TIMELINESS

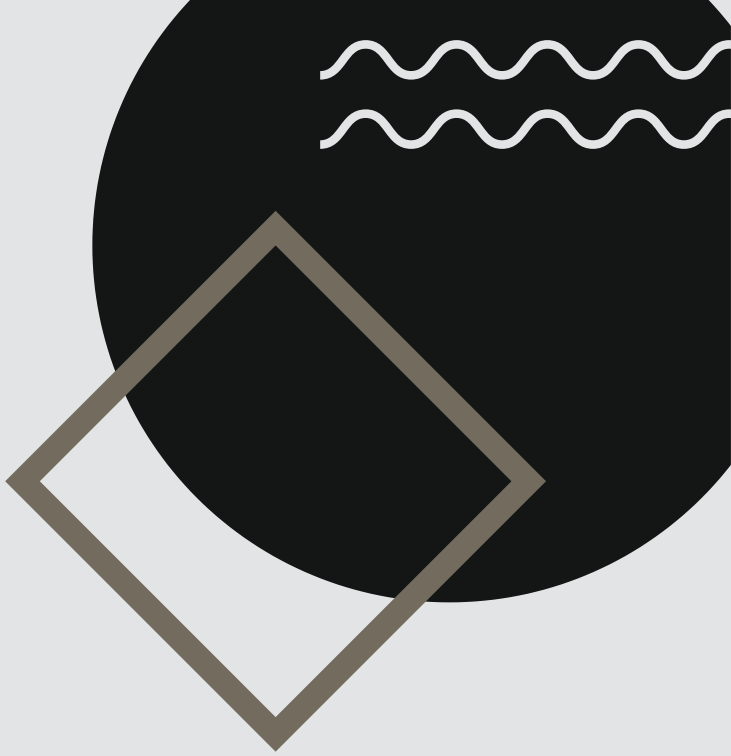


Data is uniform across systems and available when needed for decision making.



BIG DATA CONCEPTS

Big data is transforming how organizations capture, process, and act on information. Understanding the 3 Vs and the supporting technology ecosystem is key to unlocking business value.



The 3 Vs of Big Data

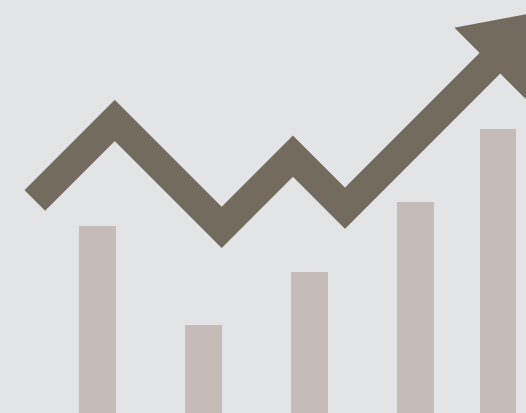
Volume: massive quantities of data from transactions, sensors, and social platforms. Velocity: data generated and processed at high speed, often in real time. Variety: structured, semi-structured, and unstructured formats.

Key Technologies

Hadoop: distributed storage and processing for large datasets. Spark: fast in-memory analytics engine. NoSQL databases: flexible storage for unstructured and semi-structured data.

Business Value

Big data enables predictive insights, personalized customer experiences, operational efficiency, and data-driven strategic decisions that create competitive advantage.



DATA WAREHOUSE VS DATA LAKE

Structured, schema-on-write storage optimized for reporting and business intelligence. Ideal for clean, consistent, and query-ready data used by analysts and decision makers.

Flexible, schema-on-read repository that stores raw data in native formats. Supports advanced analytics, machine learning, and exploratory data science at scale.

CHOOSING THE RIGHT APPROACH

MASTER DATA MANAGEMENT (MDM)

SINGLE SOURCE OF TRUTH



MDM creates a unified, authoritative master data record shared across the organization, eliminating silos and ensuring everyone relies on the same trusted data.

CORE COMPONENTS



Integration connects disparate systems, cleansing removes duplicates and errors, and governance establishes ownership, policies, and standards for master data.

KEY BENEFITS



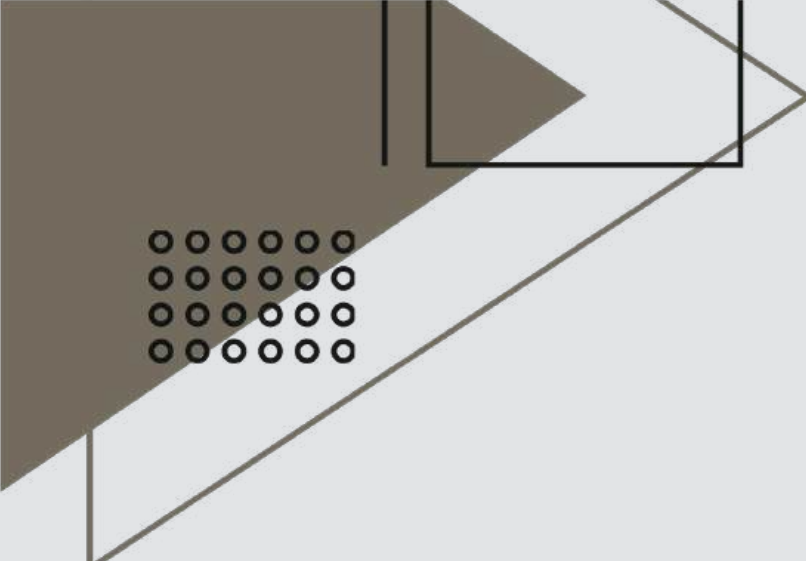
MDM improves data quality, ensures consistency across business units, reduces operational inefficiencies, and enables more reliable reporting and analytics.

MODULE 3

Introduction to Business Analyst Role
Core competencies and methodologies

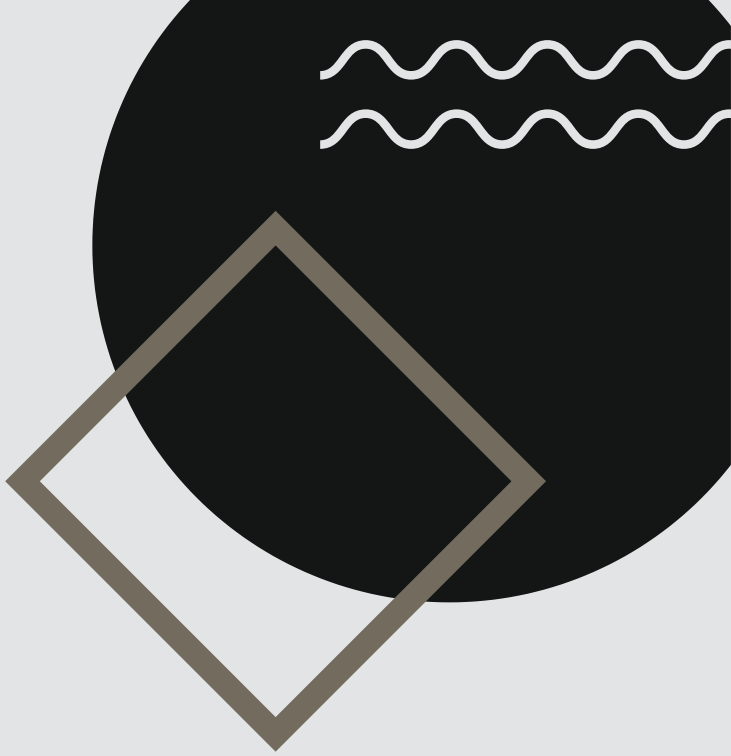
Bridging business needs and IT solutions through requirements gathering, stakeholder communication, and project support.

OBJECTIVES



BUSINESS ANALYST ROLE

Business analysts act as the critical link between business objectives and technology delivery. They gather and clarify requirements, document processes, and ensure solutions meet stakeholder needs while supporting project management, testing, and implementation.



Bridging Business & IT

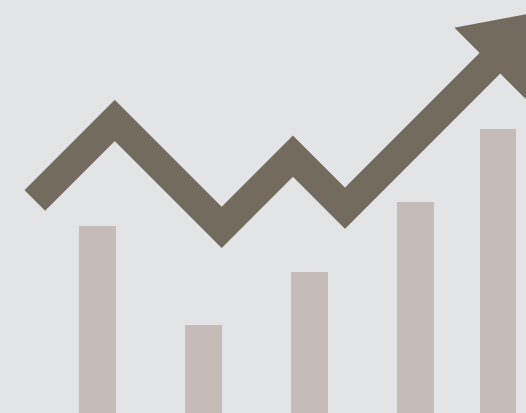
Translate business needs into IT solutions by aligning stakeholders, systems, and strategic goals.

Requirements Gathering

Elicit, document, and validate functional and non-functional requirements across the project lifecycle.

Project Support

Facilitate communication, manage expectations, and support testing, change, and project delivery.



KEY SKILLS FOR BUSINESS ANALYSTS

COMMUNICATION & INTERPERSONAL SKILLS



Active listening, clear written and verbal communication, empathy, and relationship building to align stakeholders and bridge business and IT teams.

ANALYTICAL THINKING & PROBLEM-SOLVING



Critical thinking, data interpretation, root-cause analysis, and logical reasoning to identify issues and design effective business solutions.

TECHNICAL BASICS & STAKEHOLDER MANAGEMENT



Foundational IT knowledge, requirements tools, and negotiation skills to manage expectations, resolve conflicts, and drive project success.

AGILE VS WATERFALL

Agile methodology is iterative and flexible, emphasizing collaboration, adaptive planning, and delivering value in small increments. User stories capture requirements from the end-user perspective, making it ideal for projects with changing needs.

Waterfall follows a sequential, phase-based approach with comprehensive documentation at each stage. Requirements are defined upfront, making it best suited for projects with stable scope and strict regulatory or compliance requirements.

METHODOLOGIES

STAKEHOLDER MANAGEMENT

Effective stakeholder management keeps business analysis efforts aligned with organizational priorities and drives successful project outcomes.

Identify Stakeholders

Map all internal and external parties who influence or are affected by the project, including sponsors, users, IT teams, and regulators.

Understand Needs & Expectations

Engage stakeholders to capture their goals, priorities, and concerns, ensuring requirements reflect real business needs.

Manage Conflicts & Feedback

Maintain open communication channels, address concerns early, and use feedback to refine solutions and build trust.





REQUIREMENTS ELICITATION TECHNIQUES

Business analysts use a variety of techniques to discover, clarify, and validate stakeholder requirements. Choosing the right method depends on the audience, project scope, and type of information needed.



Interviews & Workshops

- One-on-one interviews uncover detailed stakeholder needs and pain points
- Structured or semi-structured formats allow deep exploration of requirements
- Workshops bring multiple stakeholders together to align goals and brainstorm solutions
- Facilitated sessions encourage collaboration and rapid consensus building

Surveys & Questionnaires

- Questionnaires collect input from large or dispersed stakeholder groups efficiently
- Standardized questions make responses easy to compare and analyze
- Surveys can be scaled quickly across departments, customers, or user segments
- Results provide quantitative insights to prioritize requirements and validate assumptions

Document Analysis & Observation

- Document analysis reviews existing policies, reports, forms, and system records
- Helps identify current-state processes, business rules, and data requirements
- Observation lets analysts watch users in their actual work environment
- Prototyping creates early mockups to test ideas and gather feedback before full development

PROCESS MODELING: BPMN VS FLOWCHARTS

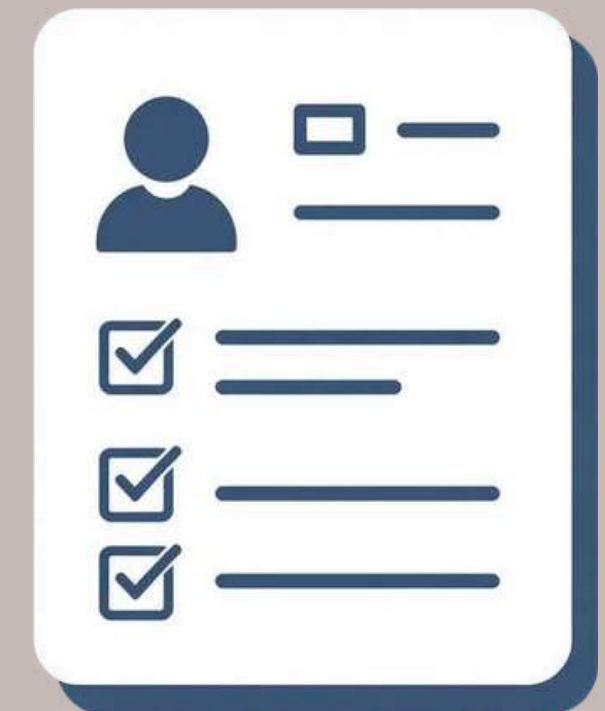
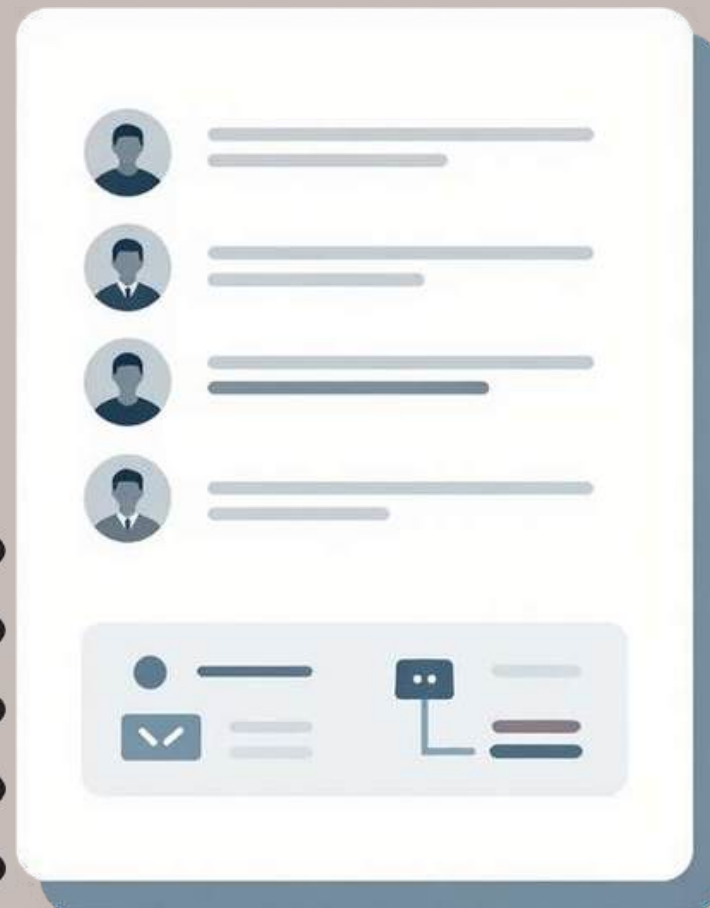
BPMN (Business Process Model and Notation) is a detailed, standardized visual language for mapping business processes. It uses specific symbols such as events, gateways, tasks, and pools to represent complex workflows, roles, and decision logic accurately across teams and systems.

Flowcharts are simple, easy-to-read diagrams that illustrate basic workflows using standard shapes like rectangles, diamonds, and arrows. They are ideal for quickly communicating processes, decision paths, and sequential steps without requiring specialized notation knowledge.

USE CASES

USE CASES & USER STORIES

Use cases capture scenario-driven interactions between users and systems, while user stories provide agile, user-focused requirements that guide iterative development.



Use Cases: scenario-driven system interactions that define actor goals and system responses.
User Stories: concise, user-focused statements following the "As a [user], I want [goal], so that [benefit]" format.

Templates: include acceptance criteria, priority, and estimated effort to support backlog refinement.

TEMPLATES & EXAMPLES

GAP ANALYSIS & FEASIBILITY

Gap Analysis compares the current state of a process or system with the desired future state. It identifies missing capabilities, process inefficiencies, and improvement opportunities. Feasibility studies then evaluate whether the proposed solutions can be implemented successfully across technical, financial, and operational dimensions.

Feasibility dimensions:

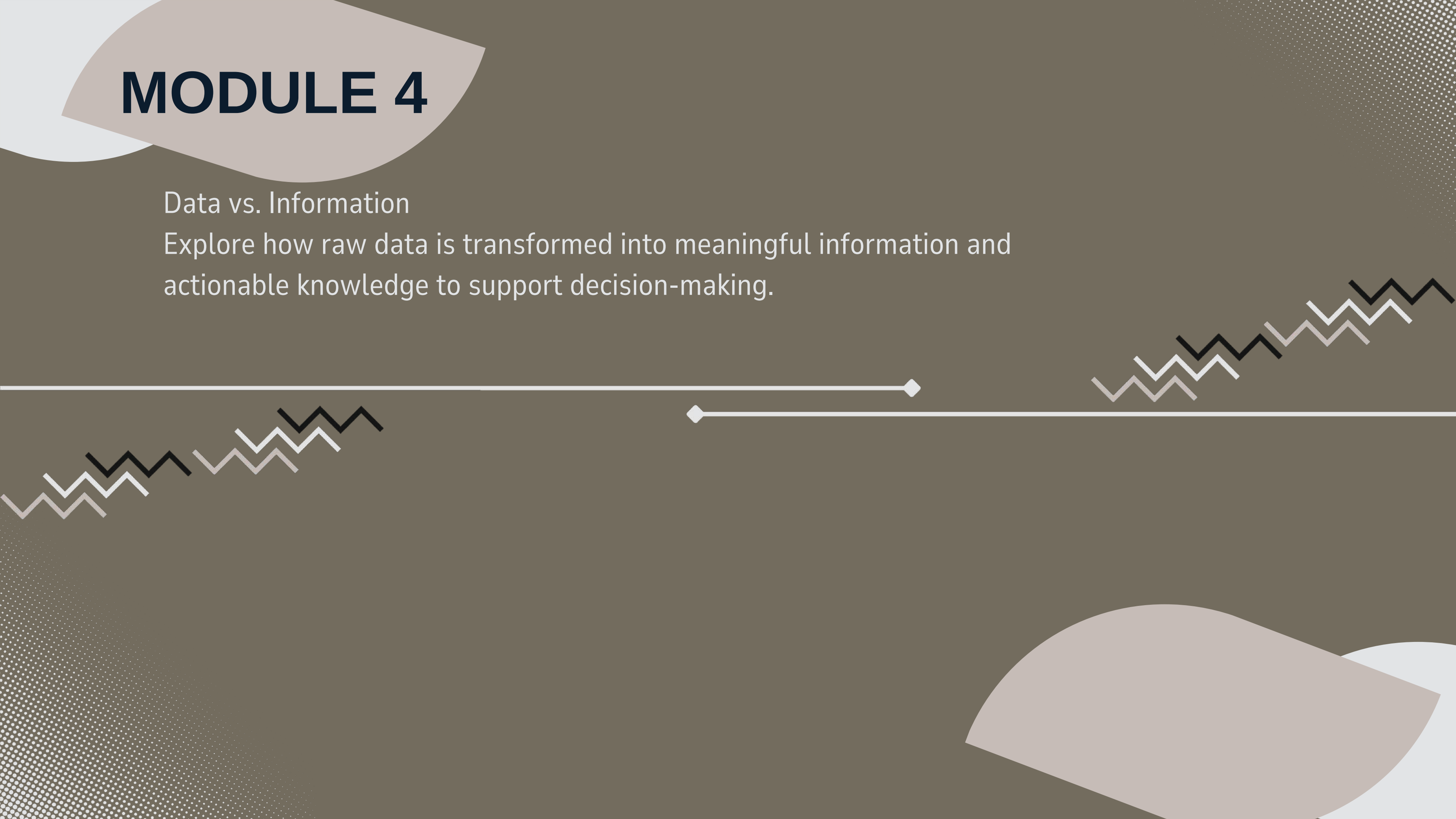
- Technical feasibility — can the solution be built with available technology and skills?
- Financial feasibility — does the expected benefit justify the cost and investment?
- Operational feasibility — will the organization adopt and sustain the solution?

Use these assessments to prioritize initiatives and reduce project risk.

MODULE 4

Data vs. Information

Explore how raw data is transformed into meaningful information and actionable knowledge to support decision-making.

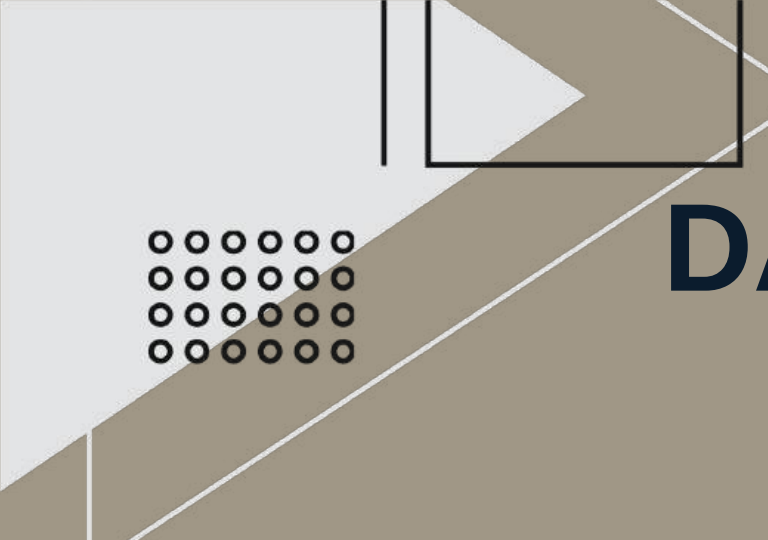


DATA VS. INFORMATION

Distinguishing raw data from processed information. Data is raw, unprocessed facts such as numbers, text, or symbols. Information is data that has been processed, organized, and interpreted to become meaningful and useful for decision making.

Examples: Raw sales figures are data; a monthly revenue report is information.
Customer names in a database are data; a ranked list of top customers is information. Temperature readings are data; a weather forecast is information.

KEY TAKEAWAY



DATA → INFORMATION → KNOWLEDGE

1

Raw Data

Unprocessed facts, figures, and signals collected from systems, sensors, and transactions. Data alone has no context or meaning until it is captured and organized.

2

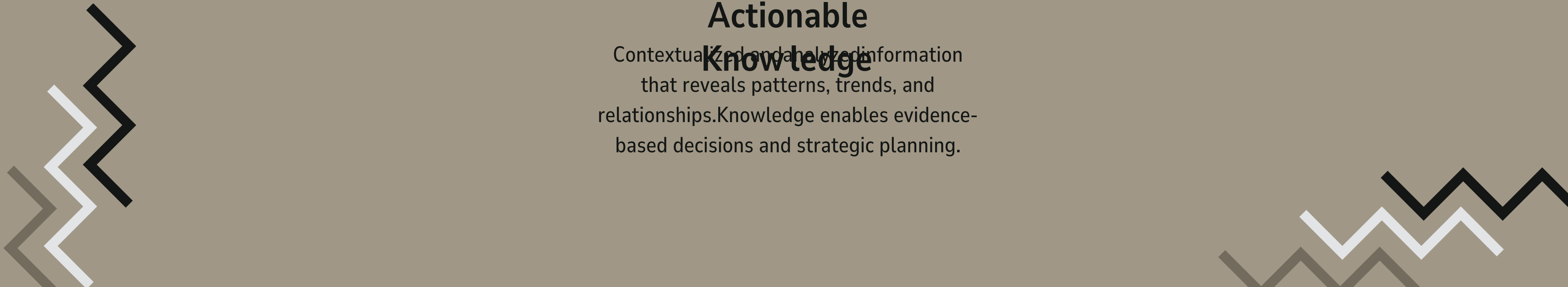
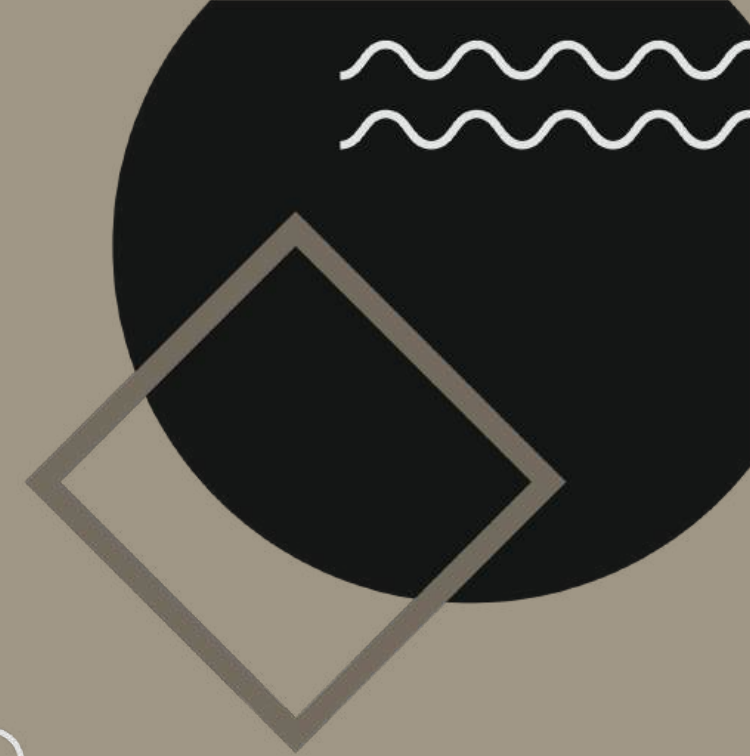
Actionable Knowledge

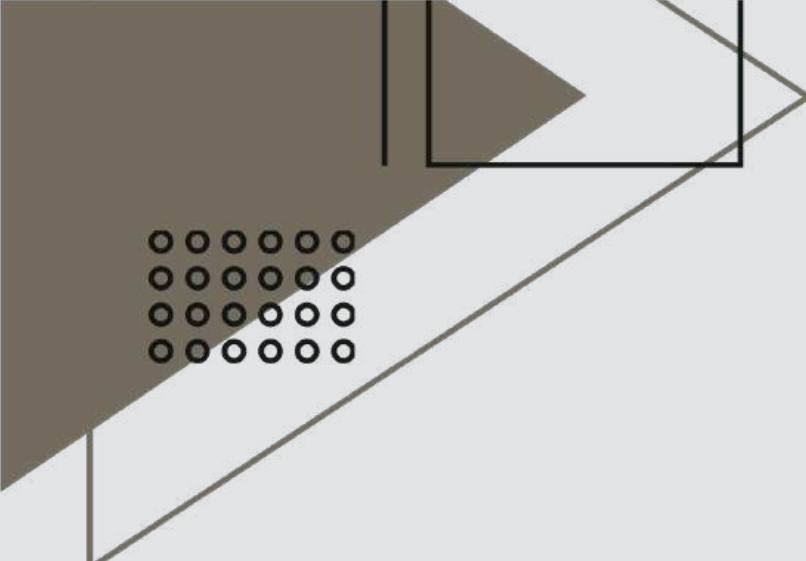
Contextualized and analyzed information that reveals patterns, trends, and relationships. Knowledge enables evidence-based decisions and strategic planning.

3

Strategic Insights

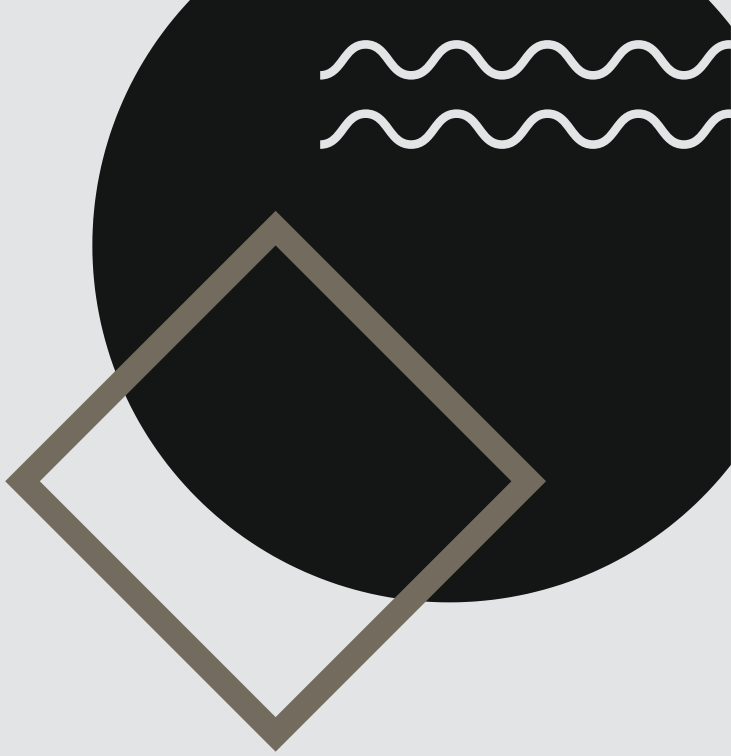
Processed, structured data with context that answers specific questions. Information supports analysis, reporting, and operational decision-making.





VALUE OF INFORMATION

Information supports long-term planning, risk reduction, and alignment with organizational goals.



Evidence-Based Decisions

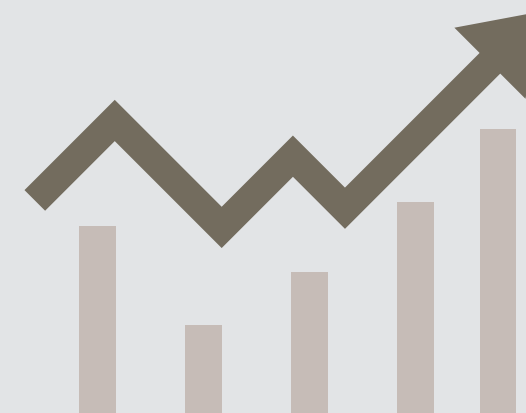
Replaces assumptions with facts, enabling objective, reliable choices grounded in real data.

Improved Accuracy & Confidence

Reduces uncertainty and errors, helping leaders forecast outcomes and act with greater confidence.

Strategic Planning Enablement

Transforms processed data into actionable insight for smarter business decisions.



ANALYTICS TYPES: DESCRIPTIVE, PREDICTIVE & PRESCRIPTIVE

DESCRIPTIVE ANALYTICS



What happened?
Analyzes past data to identify trends, patterns, and historical performance.
Example: last quarter's sales report.

PREDICTIVE ANALYTICS



What could happen?
Uses statistical models and machine learning to forecast future outcomes.
Example: demand forecasting.

PRESCRIPTIVE ANALYTICS



Prescriptive Analysis

What should we do?
Recommends optimal actions based on predictions and constraints. Example: dynamic pricing strategies.

MODULE 5

DATA-DRIVEN DECISION
MAKING, CASE
DISCUSSIONS, AND DATA

GOVERNANCE & ETHICS

Principles, Practices, and Building a Culture of Responsible Data Use
2026-07-25



OVERVIEW OF DATA-DRIVEN DECISION MAKING

WHAT IS DDDM?



Data-driven decision making (DDDM) is the practice of using data, analytics, and evidence to guide business choices rather than relying solely on intuition or experience.

WHY IT MATTERS



In modern business systems, DDDM enables organizations to respond faster to market changes, reduce uncertainty, and align operations with measurable goals.

KEY OUTCOMES



Expected outcomes include improved accuracy, faster decision cycles, better risk management, and a culture of continuous improvement and accountability.



PRINCIPLES OF DATA-DRIVEN DECISION MAKING

Core principles that guide reliable, ethical, and effective data-driven decision making across the organization.



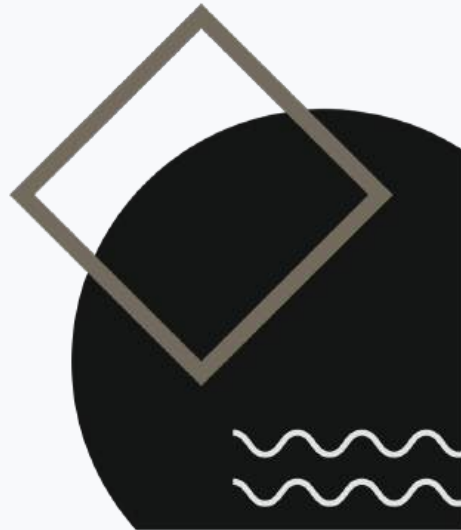
Evidence-Based Approach & Data Quality

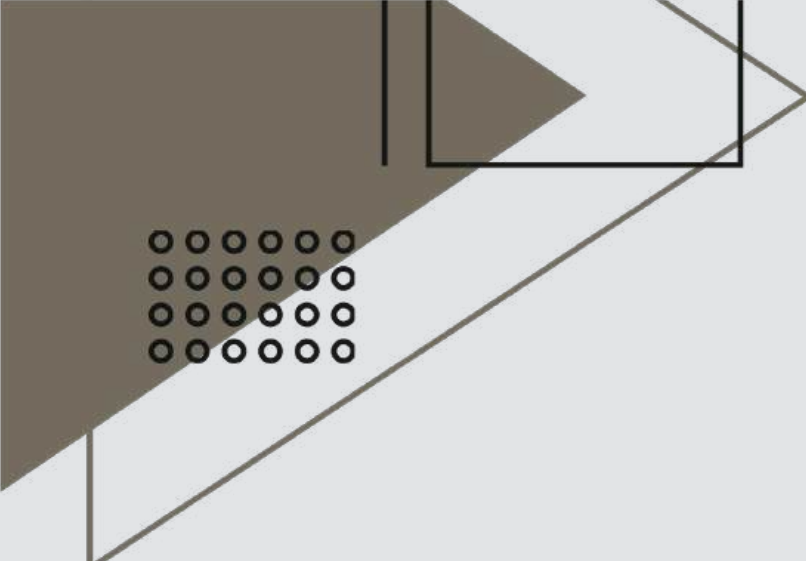
- Ground decisions in verified facts rather than assumptions or intuition alone.
- Define clear metrics and KPIs aligned with business objectives.
- Ensure data accuracy, completeness, and consistency before analysis.
- Establish validation routines and quality checks at every stage.

Transparency & Repeatability

- Document data sources, methods, and assumptions for every decision.
- Use reproducible workflows so analyses can be revisited and verified.
- Share methodologies openly with stakeholders to build trust.
- Maintain clear audit trails from raw data to final recommendation.

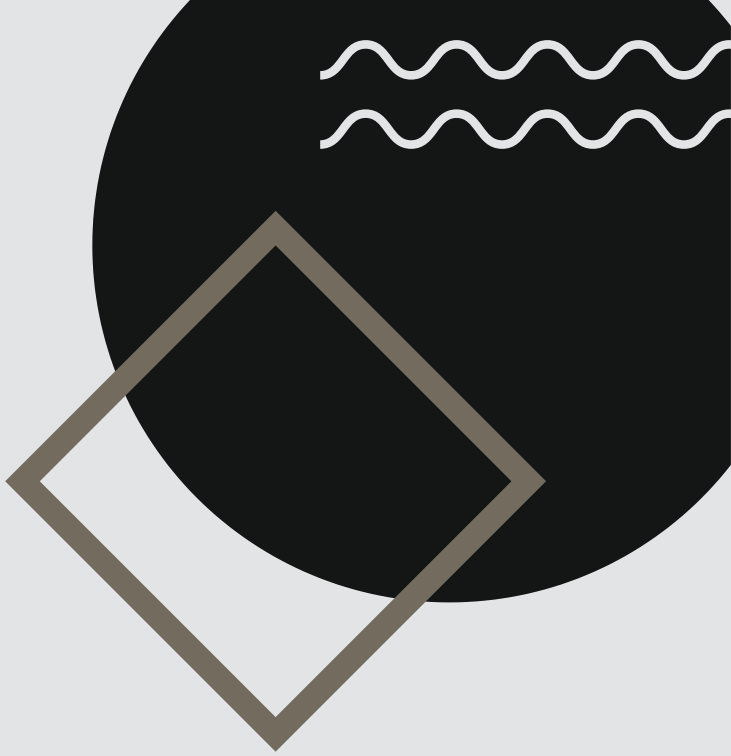
Continuous Learning & Adaptation

- Treat decision-making as an iterative process, not a one-time event.
 - Monitor outcomes and feed results back into future analyses.
 - Stay open to updating models and assumptions as new data emerges.
 - Encourage experimentation, learning from failures, and continuous improvement.
- 



BENEFITS OF DATA-DRIVEN DECISIONS

Data-driven decisionmaking transformshow organizations plan, operate, and compete. By grounding choices in evidence rather than assumptions, teams can reduce uncertainty, move faster, and align actions with strategic goals.



Improved Accuracy and Reduced Bias

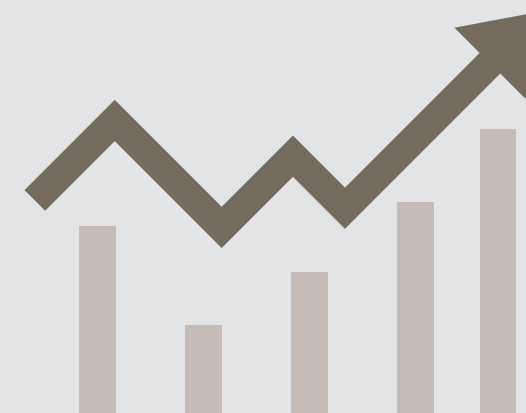
Relying on verified data reduces guesswork and counters cognitive biases, leading to more objective and reliable conclusions.

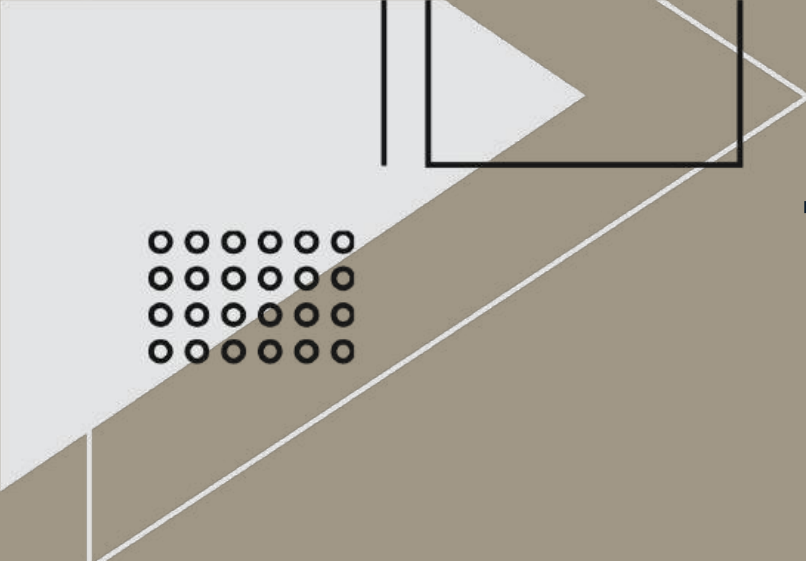
Faster and More Confident Decision Cycles

Clear, timely insights enable teams to act quickly and with confidence, shortening planning cycles and accelerating business agility.

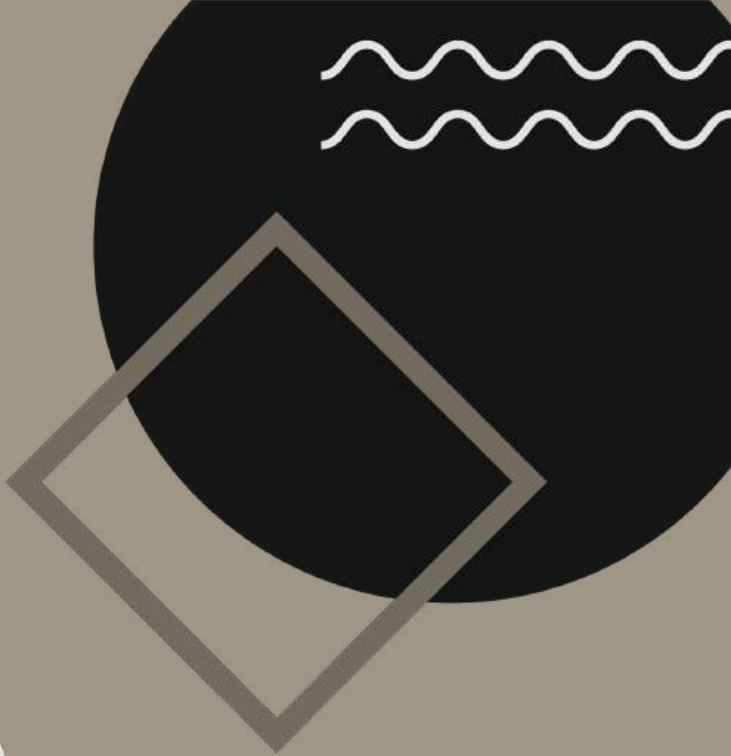
Better Risk Management and Innovation

Data helps identify risks early, uncover opportunities, and support experimentation that drives innovation while protecting the organization.





THE DATA-DRIVEN DECISION PROCESS



1

Step 1: Data Collection

Gather relevant internal and external data, then clean and validate it to ensure accuracy, completeness, and consistency before analysis begins.

2

Step 2: Analysis & Interpretation

Apply descriptive, predictive, and prescriptive methods to uncover patterns. Interpret results in context to avoid misinterpretation and surface actionable insights.

3

Step 3: Decision & Implementation

Integrate insights with business goals, formulate decisions with stakeholders, implement actions, and monitor KPIs through feedback loops for continuous improvement.



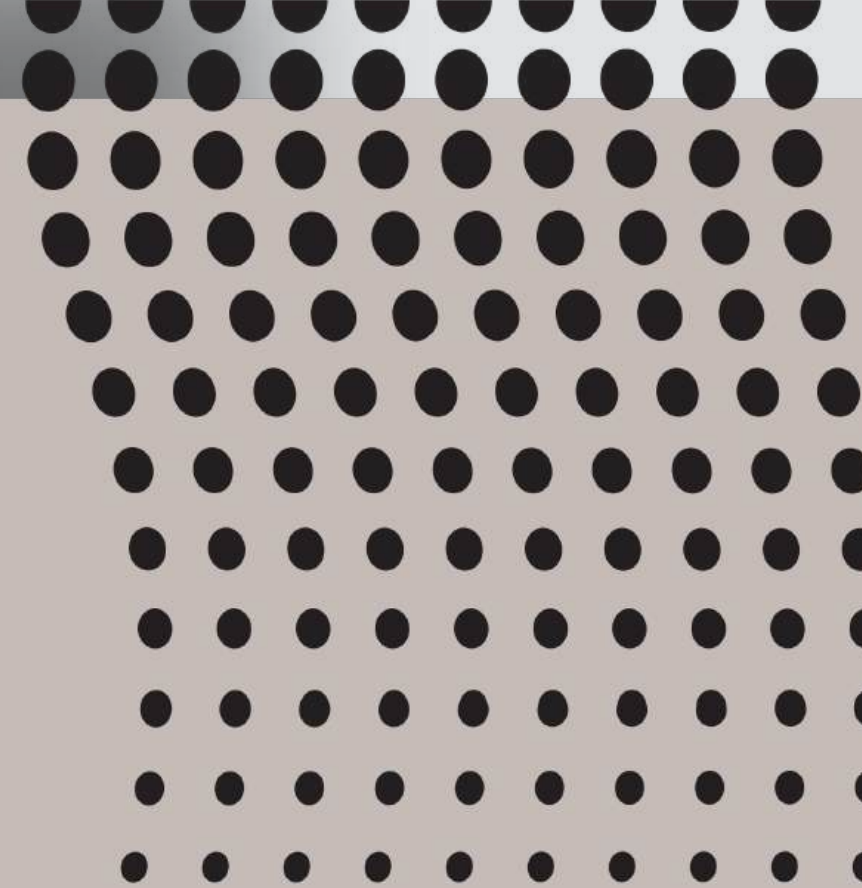
STEP 1 - DATA COLLECTION AND PREPARATION

Sources of Data

- Internal: CRM records, transaction logs, ERP systems, and operational databases.
- External: market research, public datasets, partner feeds, and social media APIs.

Data Cleaning & Validation

- Remove duplicates, fix inconsistencies, and standardize formats.
- Validate completeness, accuracy, and timeliness before analysis.
- Document assumptions and transformations to preserve traceability.



Tools and Technologies

- ETL/ELT platforms (e.g., Informatica, Talend, Apache NiFi).
- Data quality suites (e.g., Great Expectations, OpenRefine).
- Cloud warehouses (e.g., Snowflake, BigQuery, Azure Synapse).
- Collaboration notebooks (e.g., Jupyter, RStudio, Databricks).

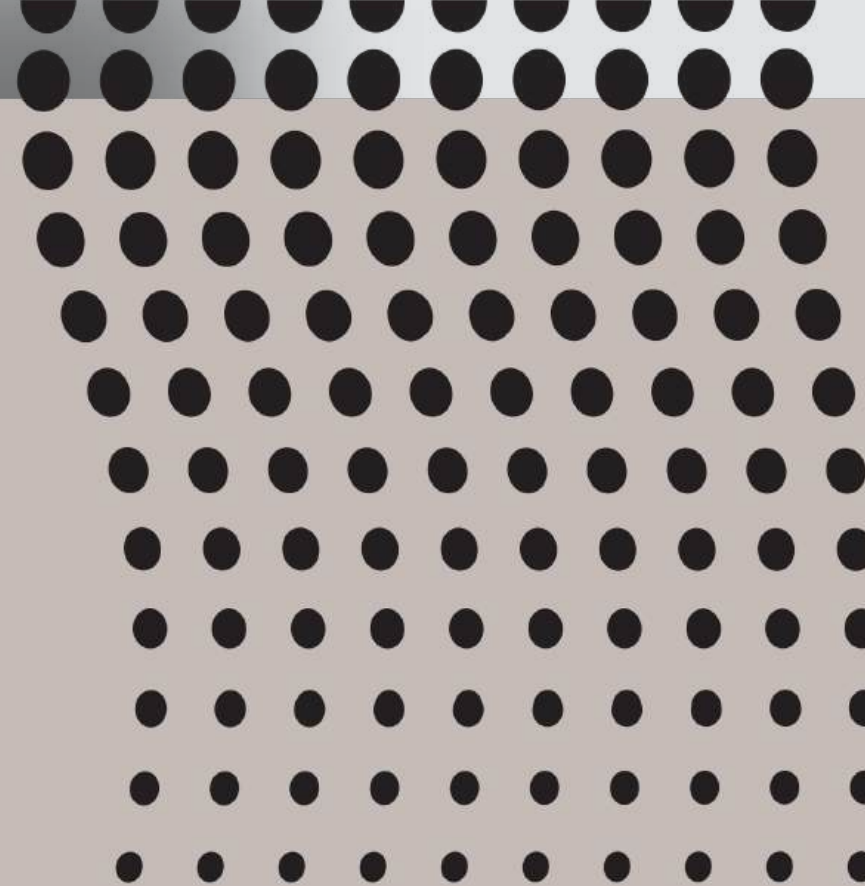
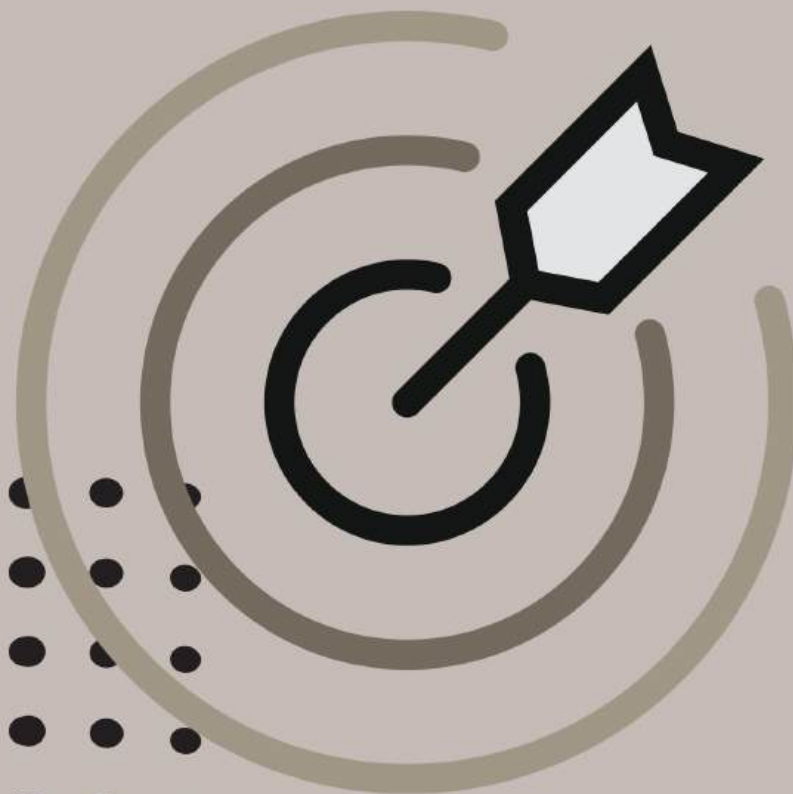
Validation Steps

1. Define data quality rules and acceptance criteria.
2. Run automated profiling and anomaly detection checks.
3. Resolve issues with data owners and revalidate.
4. Maintain audit logs for governance and reproducibility.

**FOUNDATION FOR TRUSTED
INSIGHTS**

DATA ANALYSIS AND INTERPRETATION

Analytical methods transform raw data into actionable insight. Descriptive analytics summarises what has happened, predictive analytics forecasts future outcomes, and prescriptive analytics recommends the best course of action. Choosing the right method ensures findings are relevant, reliable, and aligned with business questions.

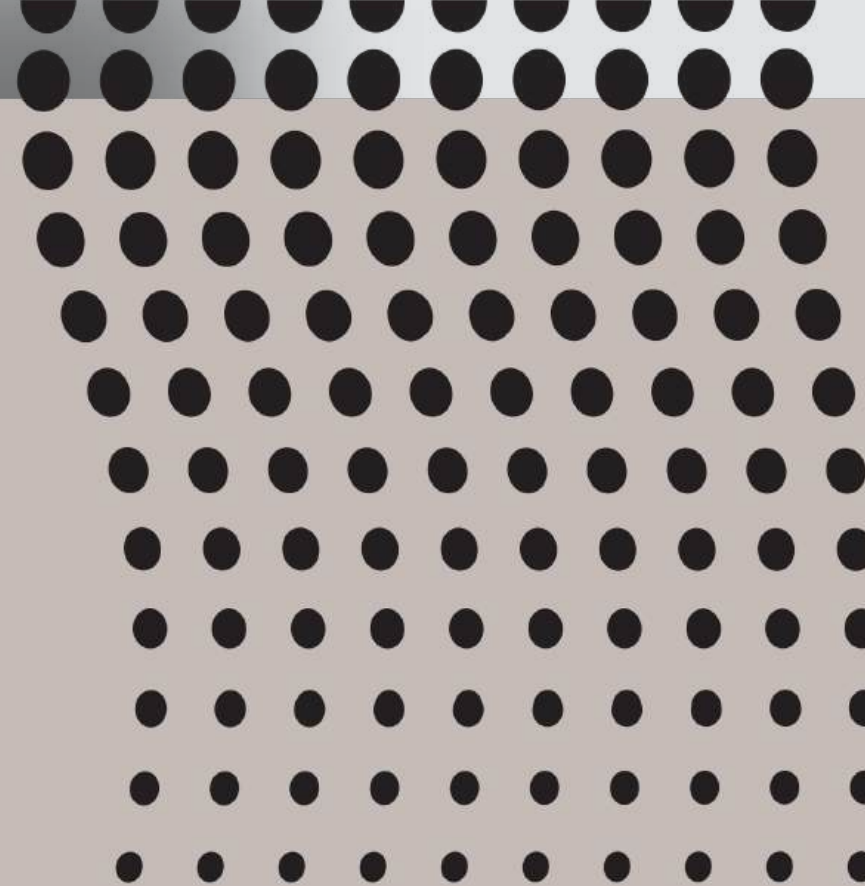
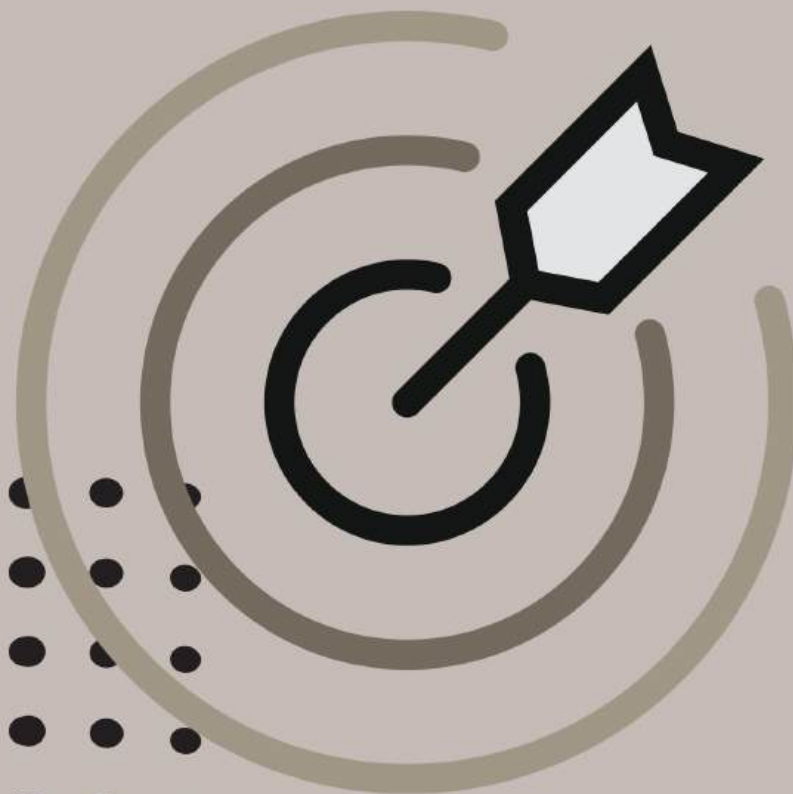


Contextual understanding is essential to avoid misinterpretation. Numbers alone rarely tell the full story; analysts must consider business conditions, data limitations, and underlying assumptions. Critical review, domain expertise, and clear communication help stakeholders draw accurate conclusions and make sound decisions.

STEP 2

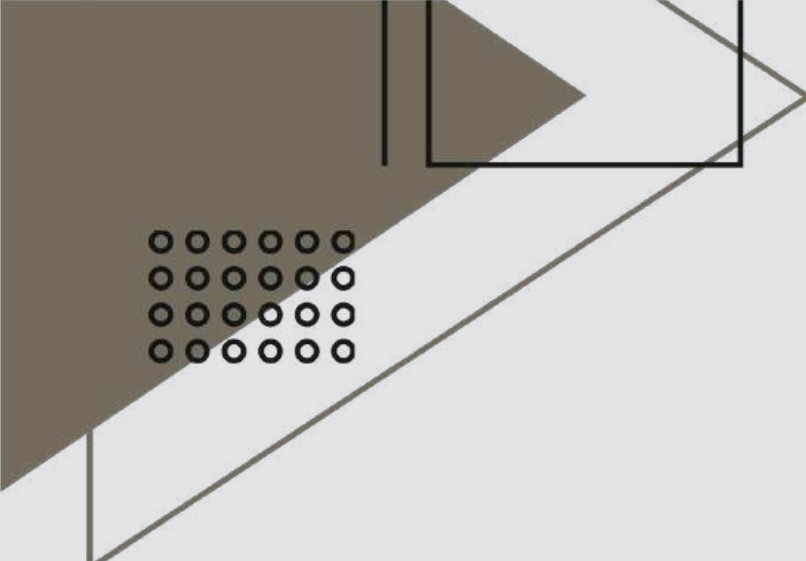
STEP 3: DECISION FORMULATION

Transform data insights into actionable decisions by aligning them with strategic business goals. Evaluate trade-offs, prioritize opportunities, and ensure recommendations are grounded in reliable evidence rather than assumptions.



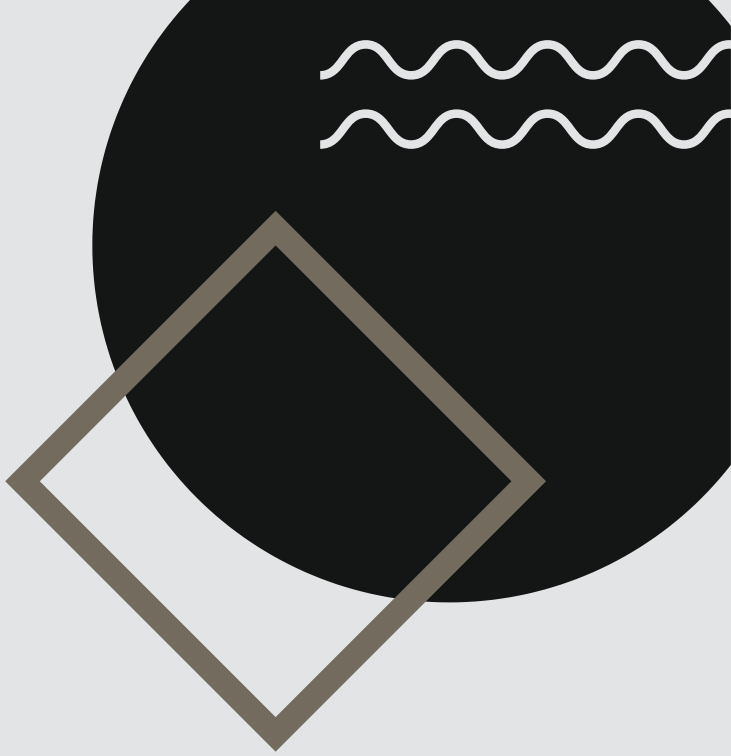
Stakeholder involvement and communication
Scenario analysis and forecasting
Iterative refinement and risk assessment

INTEGRATING INSIGHTS WITH BUSINESS GOALS



STEP 4 - IMPLEMENTATION AND MONITORING

Translate data insights into concrete actions and track their impact over time. This step ensures decisions are executed effectively and continuously refined through feedback and performance measurement.



Actioning Decisions

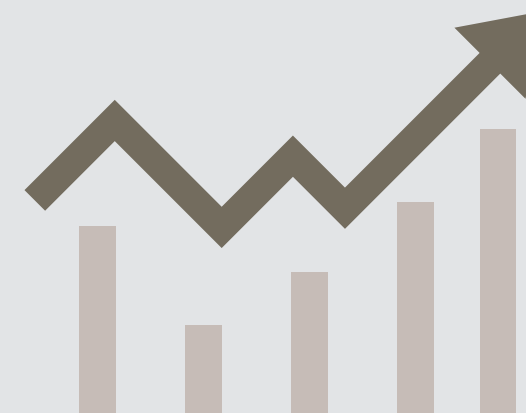
Deploy initiatives based on analytical insights, assigning clear ownership and timelines to drive execution.

KPIs and Monitoring

Define relevant KPIs and dashboards to measure impact, detect deviations, and validate expected outcomes.

Feedback Mechanisms

Establish review loops and feedback channels to learn from results and iteratively improve decisions.

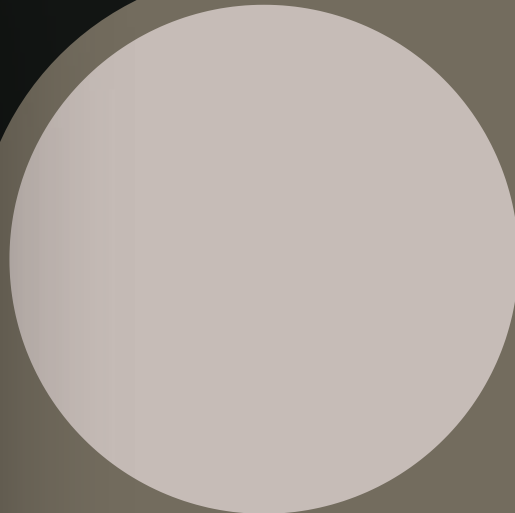


CHALLENGES AND PITFALLS

Data quality issues can undermine trust in analytics and lead to flawed conclusions. Confirmation bias may cause teams to favor data that supports existing beliefs. Overreliance on data can ignore valuable human intuition and context. Privacy concerns arise when personal or sensitive information is collected, shared, or used without proper safeguards.

Rigorous validation ensures data is accurate, complete, and fit for purpose before it drives decisions. Fostering critical review helps teams challenge assumptions and question interpretations. Balancing intuition with data brings context, experience, and creativity into the decision process. Ethical safeguards protect privacy, promote fairness, and maintain accountability.

MITIGATION STRATEGIES

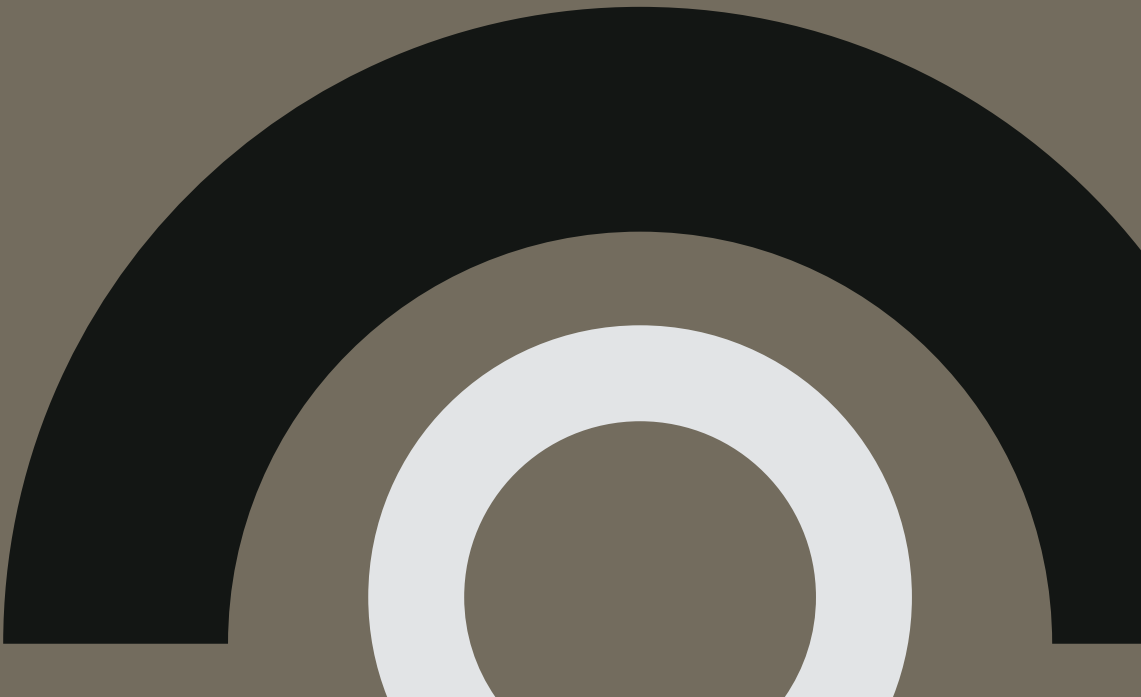


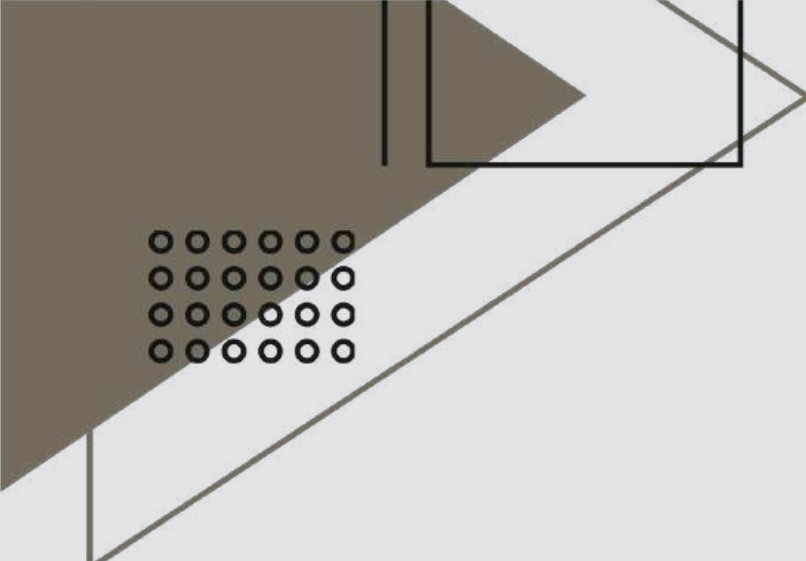
MODULE 6

CASE

DISCUSSIONS

Real-Life Business Scenarios





BUSINESS SCENARIO

1: CUSTOMER CHURN REDUCTION

A telecom company was experiencing high customer churn and needed a data-driven approach to improve retention. By analyzing customer behavior, support interactions, and feedback, the company identified at-risk subscribers and launched targeted retention campaigns.



Context

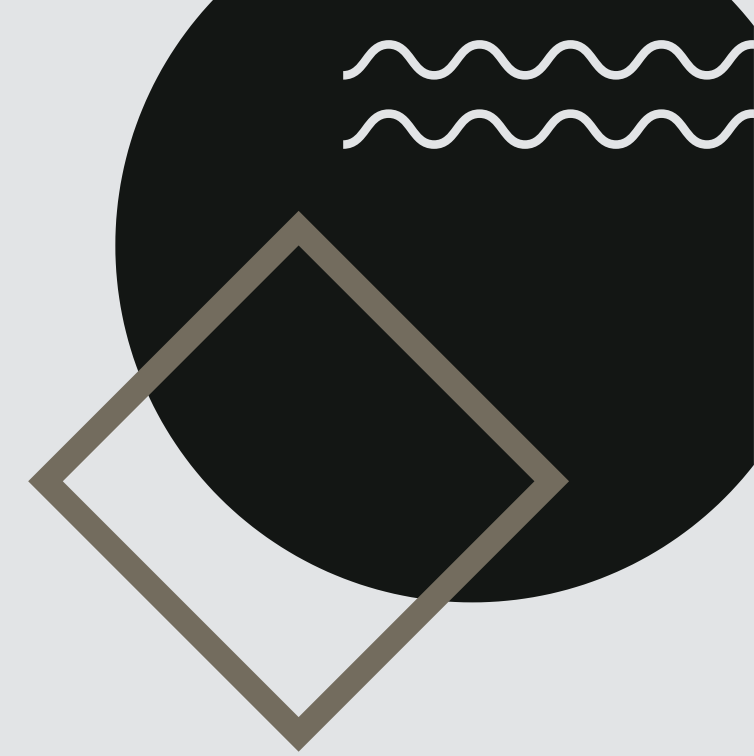
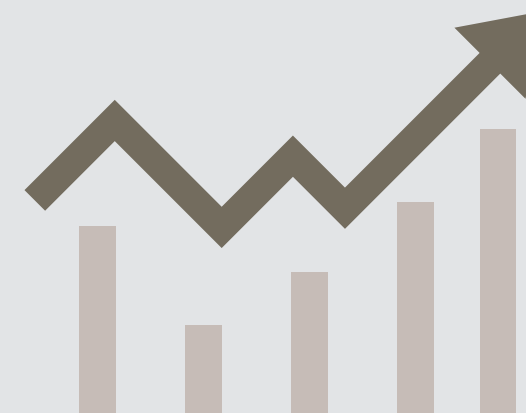
Telecom company facing high customer churn and declining retention rates.

Data Sources Used

Customer usage patterns, support tickets, and survey responses.

Outcome

Targeted retention campaigns delivered a 15% reduction in churn.



SCENARIO ANALYSIS: APPROACH AND INSIGHTS

DATA AGGREGATION AND SEGMENTATION



Consolidated customer data from usage logs, support tickets, and survey responses to create meaningful behavioral segments.

PREDICTIVE MODELING OF CHURN RISK

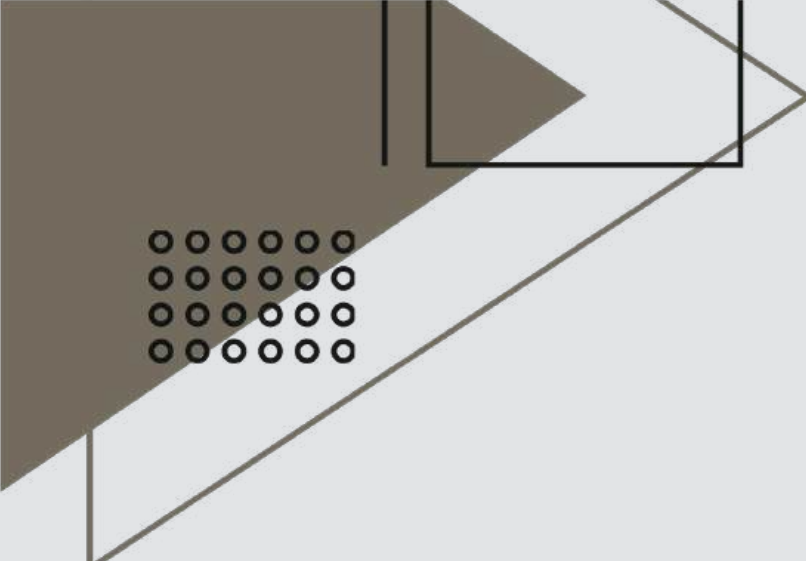


Applied descriptive and predictive analytics to identify at-risk customers and forecast churn probability.

TAILORED INTERVENTION STRATEGIES

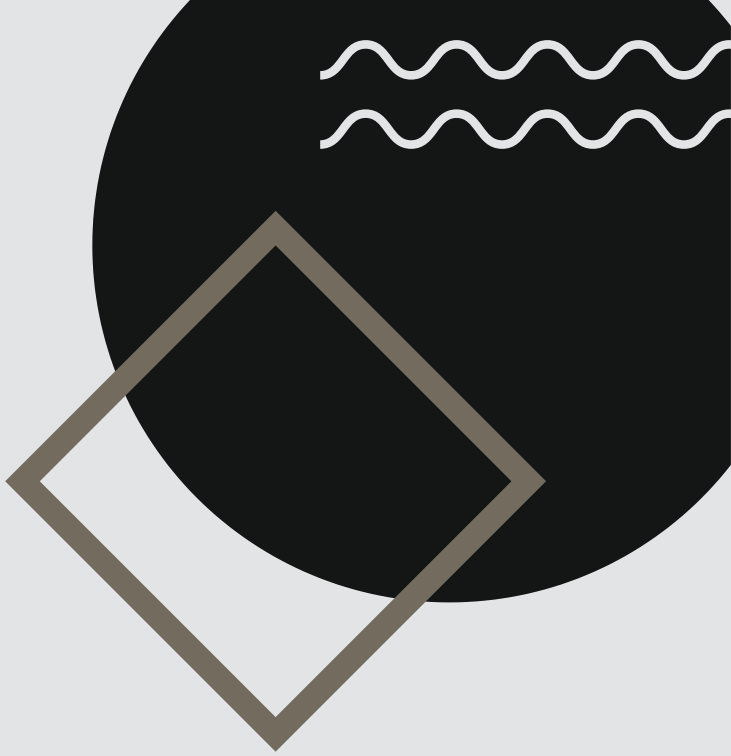


Designed targeted retention campaigns and personalized offers based on risk scores and segment needs.



SCENARIO 2: SUPPLY CHAIN OPTIMIZATION

A retail client optimized inventory levels by analyzing sales trends, supplier lead times, and logistics data. The data-driven approach resulted in a 20% reduction in stockouts and a 10% reduction in supply chain costs.



Context

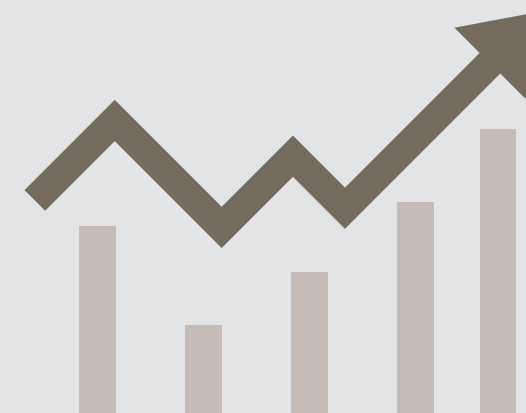
Retail client seeking to optimize inventory levels across multiple locations and reduce operational costs.

Data Sources

Sales trends, supplier lead times, logistics and transportation data, and warehouse inventory levels.

Results

20% reduction in stockouts and 10% cost savings through improved demand forecasting and supplier coordination.



LESSONS LEARNED FROM CASE DISCUSSIONS

CROSS- FUNCTIONAL COLLABORATION



Break down silos between data, operations, marketing, and IT teams to align insights with business action.

CLEAN, TIMELY DATA



Reliable decisions require accurate, up-to-date data with consistent validation and governance practices.

ITERATIVE TESTING & LEARNING



Treat decisions as experiments: measure outcomes, refine models, and apply learnings continuously.

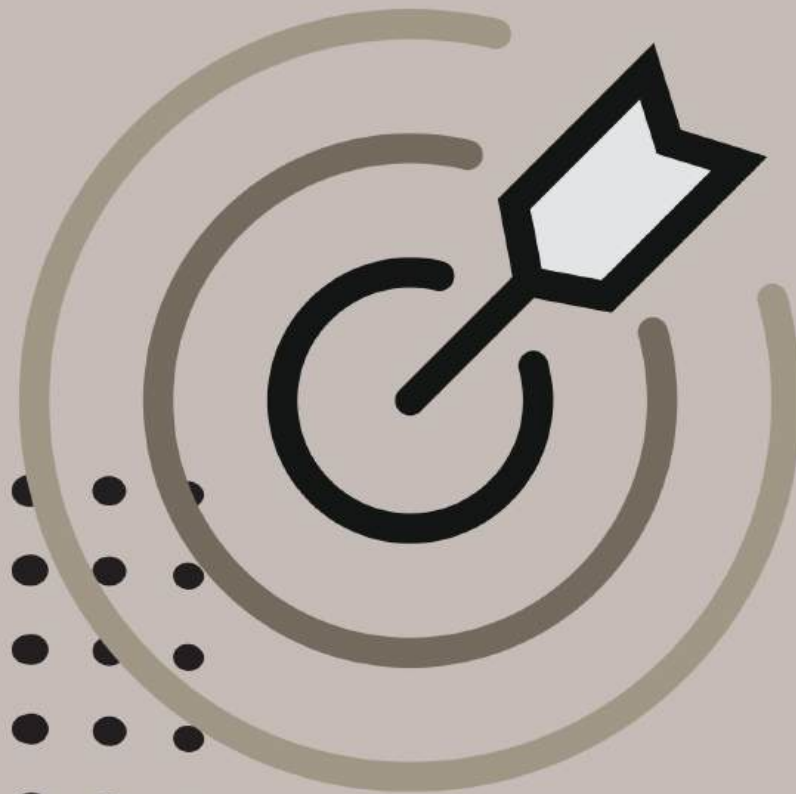


DATA GOVERNANCE & ETHICS

2026-07-25

WHAT IS DATA GOVERNANCE?

Data governance is the formal framework of policies, procedures, and standards that guide how data is managed across an organization. It defines who is accountable for data assets, how data is collected, stored, accessed, and maintained, and the rules that ensure it remains reliable, secure, and aligned with business objectives.



Quality: establish standards for accuracy, completeness, and consistency.

Security: protect data from unauthorized access, breaches, and loss.

Compliance: meet regulatory requirements and internal audit expectations.

ENSURING QUALITY, SECURITY, AND COMPLIANCE



MODULE 7

DATA GOVERNANCE POLICIES AND PROCEDURES

Core policy areas that ensure data is managed securely, consistently, and in compliance with organizational and regulatory standards.



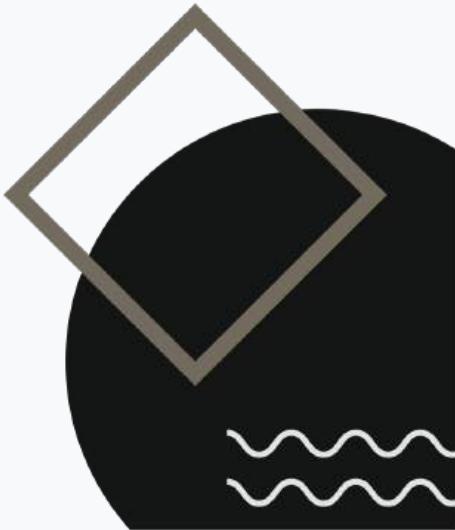
Data Access Controls & Lifecycle Management

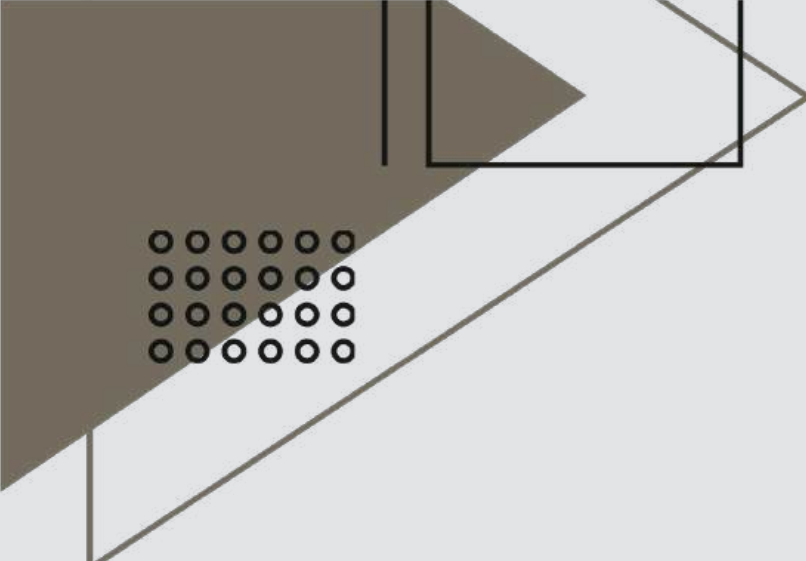
- Define role-based access controls to limit data exposure to authorized users
- Implement authentication, authorization, and least-privilege principles
- Manage data from creation and storage through archival and secure disposal
- Maintain metadata, lineage, and retention schedules across the data lifecycle

Incident Response & Audit Trails

- Establish documented procedures for detecting and responding to data incidents
- Maintain comprehensive audit trails of data access, changes, and processing activities
- Define escalation paths, containment steps, and recovery protocols
- Conduct post-incident reviews to strengthen controls and prevent recurrence

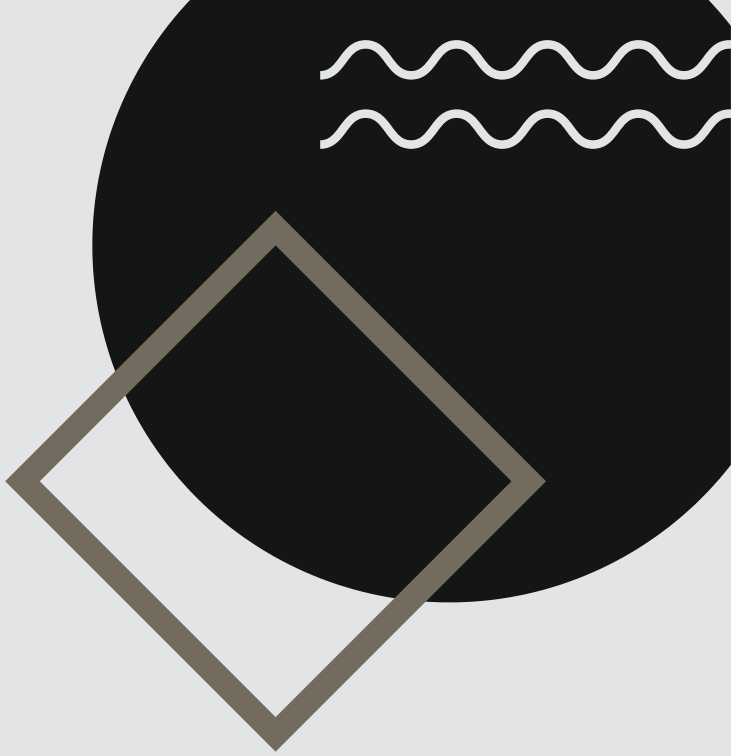
Compliance Monitoring & Enforcement

- Monitor adherence to internal policies and external regulations continuously
 - Perform regular compliance assessments, risk reviews, and control testing
 - Enforce accountability through clear roles, penalties, and remediation workflows
 - Report compliance status to leadership and adjust policies as requirements evolve
- 



ROLES AND RESPONSIBILITIES IN DATA GOVERNANCE

Clear governance roles ensure accountability across the data lifecycle, from ownership and stewardship to custody and compliance oversight.



Data Owners and Data Stewards

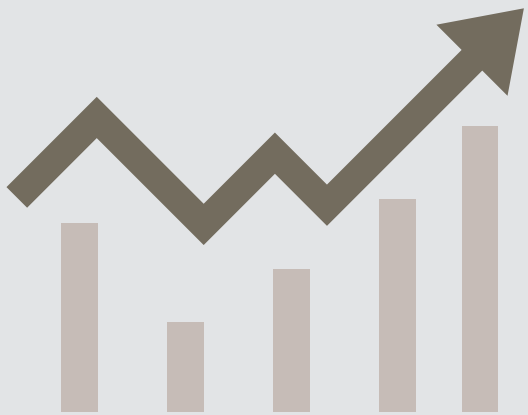
Accountable for data quality, classification, and business-aligned usage policies.

Data Custodians

Manage technical storage, security, backups, and access controls for data assets.

Compliance Officers

Oversee regulatory compliance, risk assessments, and ethical data handling standards.





PRINCIPLES OF DATA ETHICS

Guiding principles for responsible data use in
governance and decision-making.



Privacy & Confidentiality

- Protect personal and sensitive data from unauthorized access
- Apply encryption, anonymization, and access controls
- Train employees on handling confidential information responsibly

Fairness & Impartiality

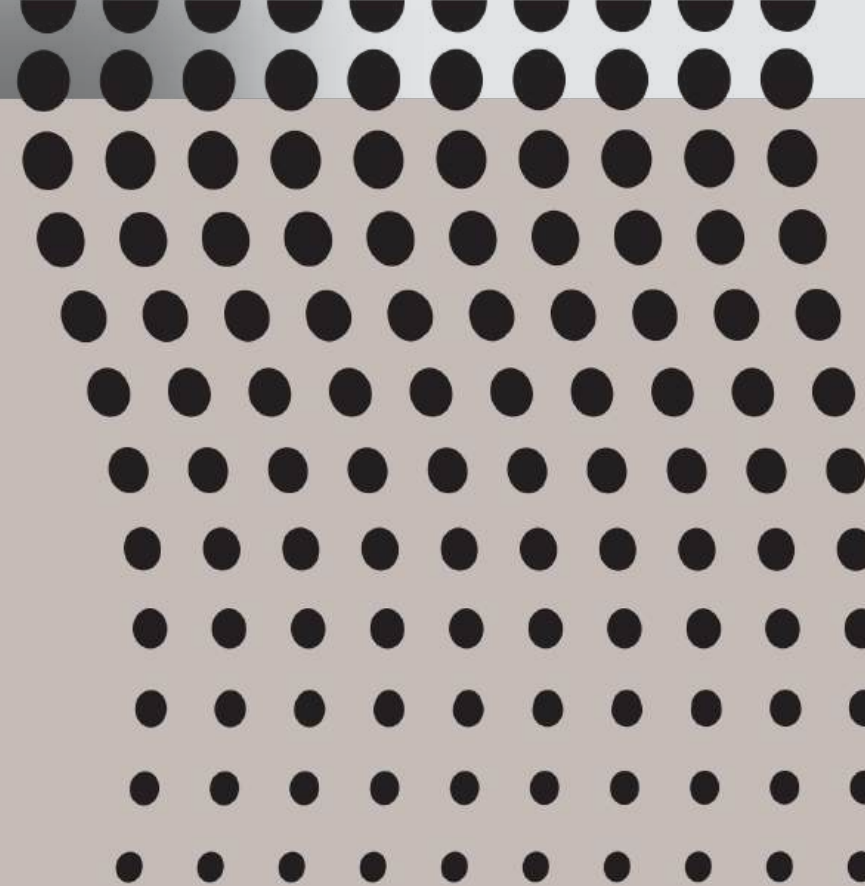
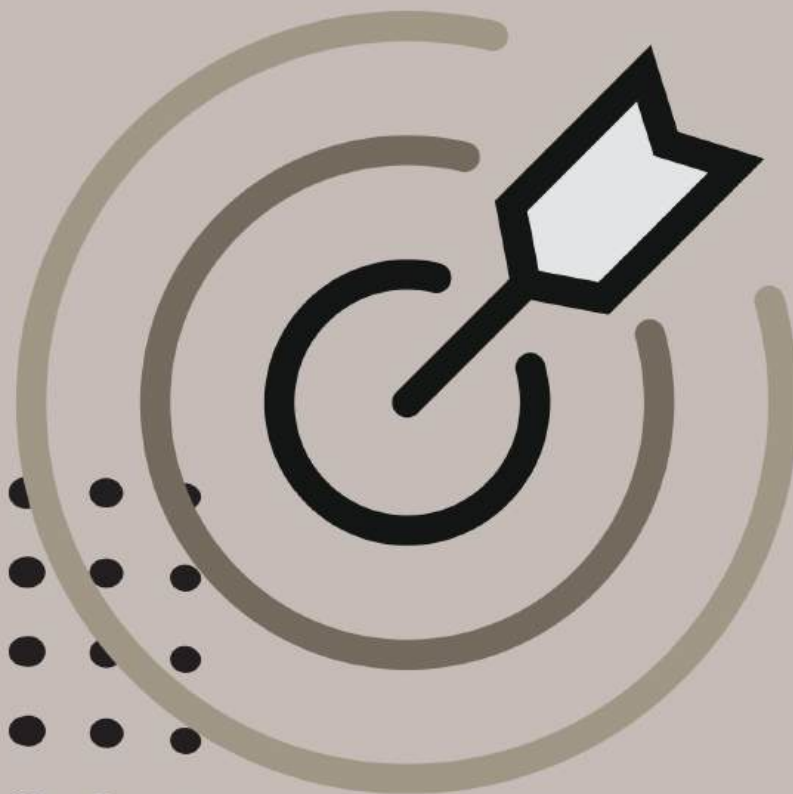
- Ensure data practices do not discriminate or reinforce bias
- Validate models and datasets for representative outcomes
- Promote equitable decision-making across all stakeholder groups

Transparency & Accountability

- Communicate clearly how data is collected, used, and shared
- Maintain audit trails and assign ownership for data decisions
- Establish channels to report and address ethical concerns

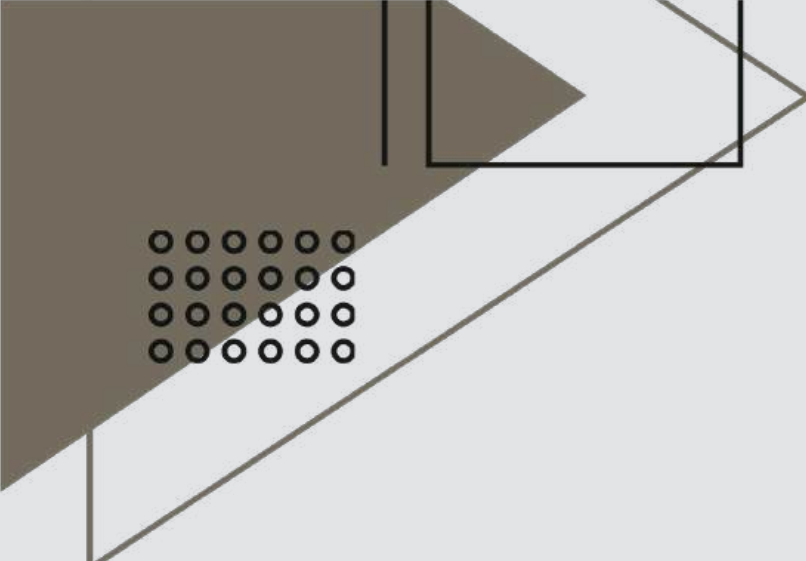
PRIVACY AND CONFIDENTIALITY

Protecting personal and sensitive data is a cornerstone of responsible data governance. Organizations must establish clear policies for identifying, classifying, and handling confidential information. This includes limiting access to authorized users, applying need-to-know principles, and maintaining secure storage practices. Regular risk assessments and data mapping help ensure that privacy controls keep pace with evolving threats and regulatory expectations.



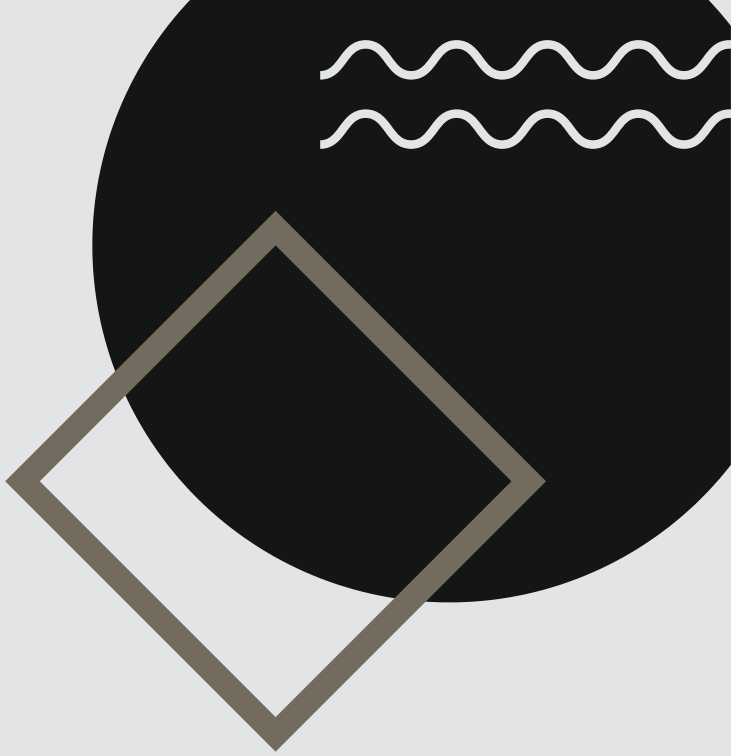
Encryption and anonymization techniques safeguard data both in transit and at rest. Strong access controls, tokenization, and pseudonymization reduce exposure while preserving utility. Employee training and awareness programs build a privacy-conscious culture, ensuring teams understand their responsibilities, recognize potential risks, and respond appropriately to breaches or policy violations.

OUR COMMITMENT



TRANSPARENCY AND ACCOUNTABILITY

Transparency and accountability are essential to building trust in data-driven systems. Organizations should make data practices visible and understandable, ensure every step of the data lifecycle can be reviewed, and empower individuals to report concerns without fear of retaliation.



Clear Data Usage Disclosures

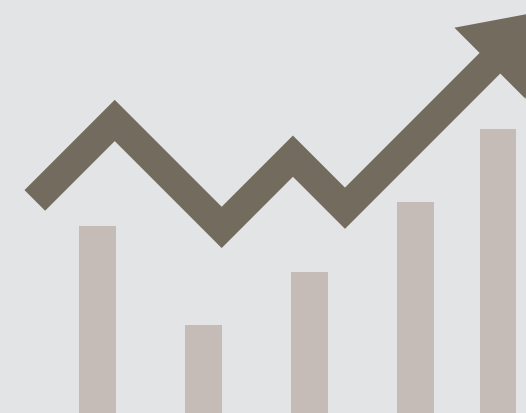
Communicate openly about what data is collected, how it is used, and who can access it.

Auditability of Data Processes

Maintain complete, traceable records of data workflows, decisions, and changes to ensure accountability.

Reporting Mechanisms for Ethical Concerns

Provide accessible, confidential channels for stakeholders to raise and resolve data ethics issues.



REGULATORY COMPLIANCE OVERVIEW

Adhering to data protection laws is essential for organizations that collect, process, and store personal information. Compliance builds trust with customers and stakeholders, reduces legal and financial risk, and demonstrates responsible data stewardship. Regulations establish clear obligations for how data must be handled, secured, and shared, helping organizations avoid costly breaches, penalties, and reputational damage.

This presentation focuses on two major regulatory frameworks: the General Data Protection Regulation (GDPR) in the European Union, and the California Consumer Privacy Act (CCPA) in the United States. GDPR emphasizes data subject rights, lawful processing, and accountability. CCPA grants consumers rights to know, delete, and opt-out of the sale of their personal information. Both frameworks shape how modern businesses design governance, privacy, and ethics programs.



GDPR KEY REQUIREMENTS

Core obligations under the General Data Protection Regulation covering individual rights, lawful consent, impact assessments, and breach response timelines.



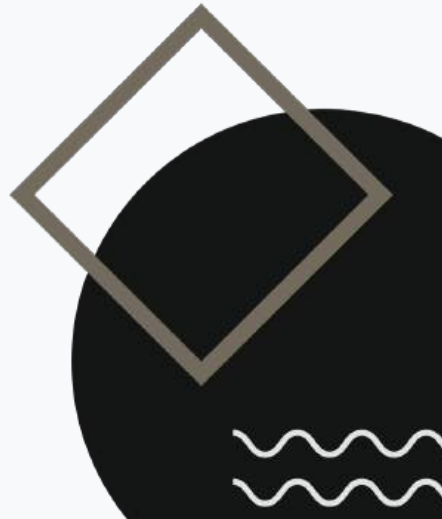
Data Subject Rights

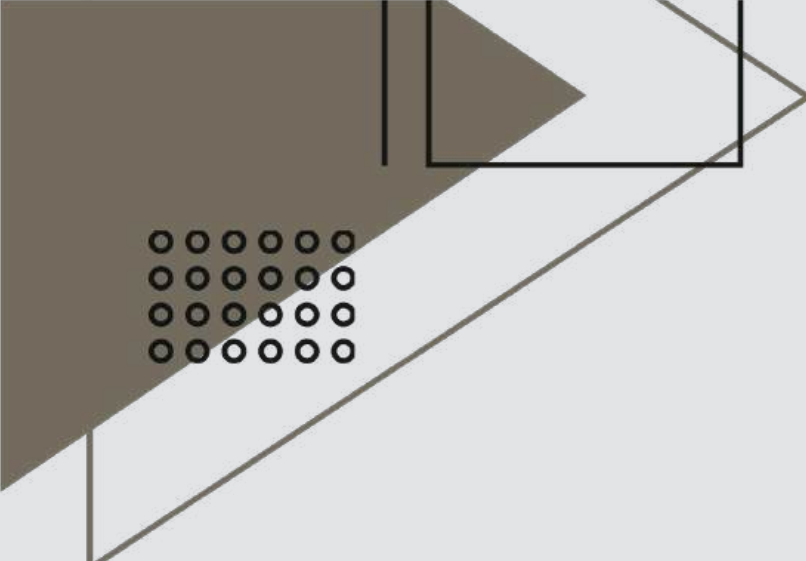
- Right of access: individuals can request copies of their personal data.
- Right to erasure ('right to be forgotten'): data must be deleted under lawful conditions.
- Right to data portability: individuals can transfer data between providers.
- Rights to rectification and restriction help ensure accuracy and control.

Consent Management & DPIAs

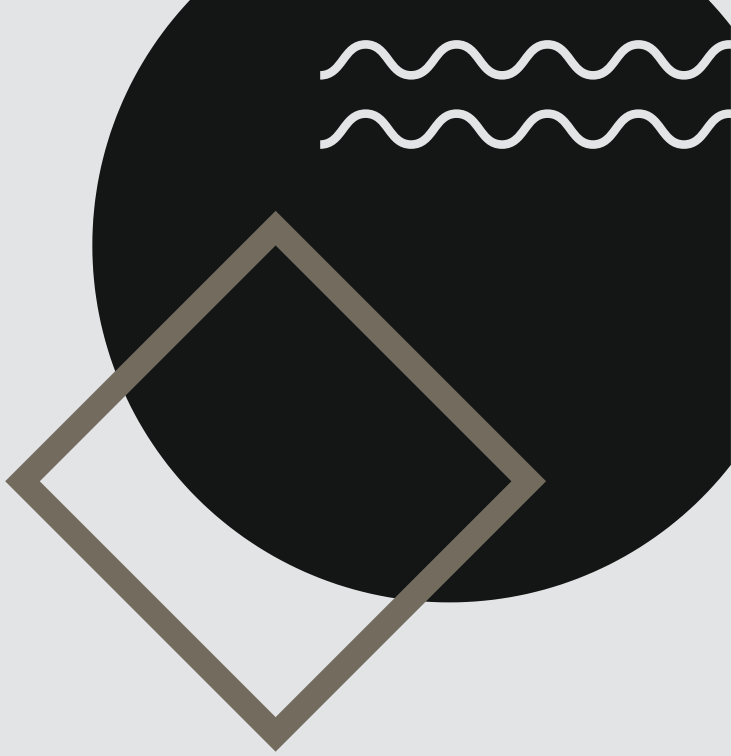
- Consent must be freely given, specific, informed, and unambiguous.
- Organizations must maintain clear records of consent and withdrawal.
- Data Protection Impact Assessments (DPIAs) identify and mitigate privacy risks.
- DPIAs are required for high-risk processing involving sensitive data or profiling.

Breach Notification

- Personal data breaches must be reported to supervisory authorities within 72 hours.
 - Individuals must be notified without undue delay if high risk of harm exists.
 - Breach records must document facts, effects, and remediation actions taken.
 - Timely reporting builds trust and demonstrates accountability.
- 



CCPA KEY REQUIREMENTS



Consumer Rights

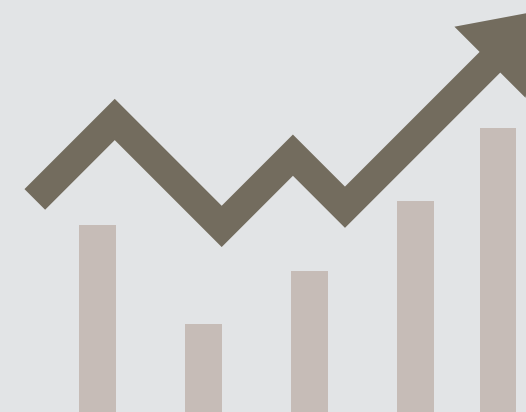
Consumers have the right to know what personal data is collected, the right to delete their data, and the right to opt-out of its sale.

Business Obligations and Penalties

Businesses must disclose data practices, respond to consumer requests promptly, and face penalties for non-compliance or data breaches.

Comparison with GDPR

CCPA and GDPR both emphasize transparency and individual rights, but differ in scope, consent requirements, and enforcement mechanisms.





BUILDING A CULTURE OF RESPONSIBLE DATA USE

Leadership commitment and role modeling set the tone

for responsible data practices. Ongoing training programs build awareness and skills across teams. Embedding ethics into data projects and encouraging open dialogue on data concerns help organizations make trustworthy, values-driven decisions.



Leadership Commitment

Senior leaders model responsible data behavior by integrating ethics into strategy, decisions, and communications. Visible commitment reinforces accountability and signals that data responsibility is an organizational priority.

Ongoing Training Programs

Regular training equips employees with practical knowledge of governance policies, privacy practices, and ethical decision-making. Continuous learning keeps teams prepared as regulations, technologies, and risks evolve.

Ethics and Open Dialogue

Ethics should be embedded into every data project through clear guidelines, review checkpoints, and risk assessments. Creating channels for open dialogue enables teams to raise concerns and collaborate on responsible solutions.

TOOLS AND TECHNOLOGIES SUPPORTING GOVERNANCE & ETHICS

DATA CATALOGUES & METADATA MANAGEMENT



Centralize data assets with searchable catalogues, maintain accurate metadata, and ensure teams can discover, understand, and trust the data they use.

PRIVACY MANAGEMENT SOFTWARE

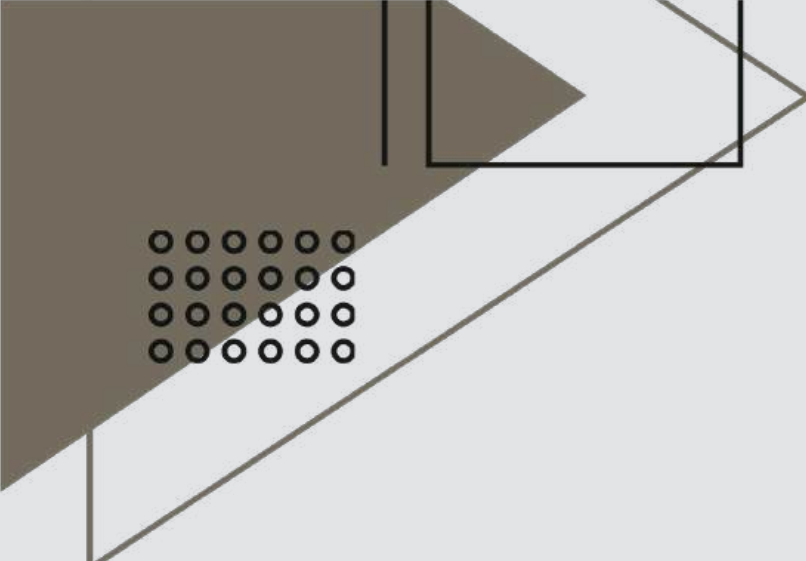


Automate privacy workflows, manage consent and data subject requests, and embed privacy-by-design principles across the data lifecycle.

COMPLIANCE TRACKING SYSTEMS

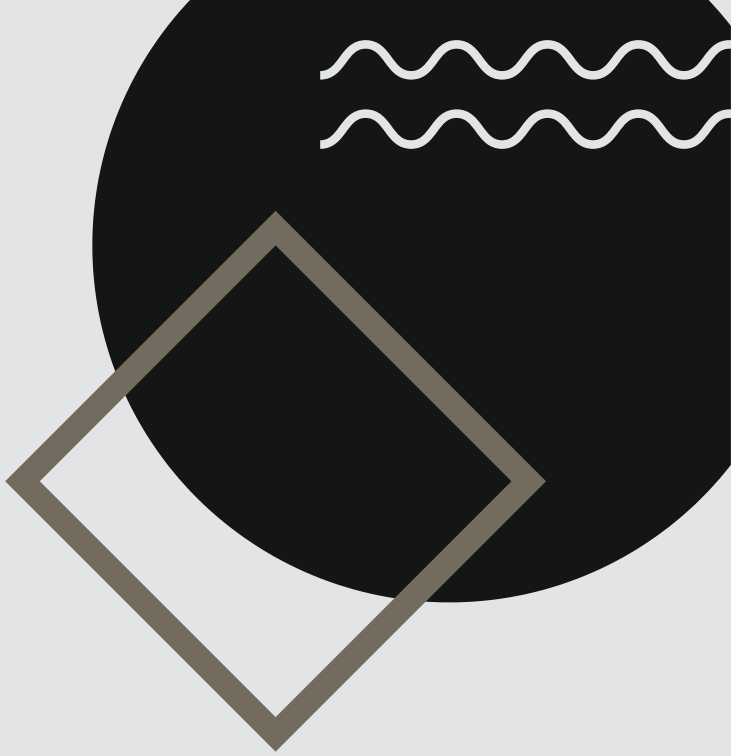


Monitor regulatory obligations, track policy adherence, manage audits, and generate evidence to demonstrate compliance with GDPR, CCPA, and other standards.



KEY TAKEAWAYS

Data-driven decision making enhances business outcomes, real-life cases exemplify practical application, and strong governance, ethics, and culture are essential for responsible data use.



Enhanced Business Outcomes

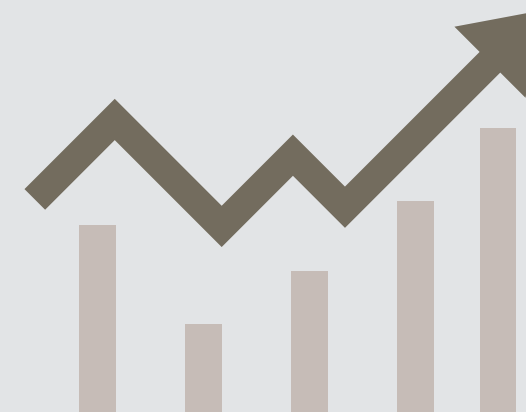
DDDM improves accuracy, speed, agility, and risk management across business systems.

Practical Application

Real-life cases such as churn reduction and supply chain optimization demonstrate data-driven value.

Governance, Ethics & Culture

Strong frameworks, compliance, and a culture of responsible use are the foundation for sustainable data practices.



NEXT STEPS AND RECOMMENDATIONS

ASSESS GOVERNANCE MATURITY



Evaluate current data governance maturity, identify gaps, and implement targeted training programs on ethics and compliance.

INTEGRATE DDDM FRAMEWORKS



Embed data-driven decision-making frameworks into business processes to improve accuracy, speed, and accountability.

PLAN REGULAR AUDITS



Schedule regular audits and governance updates to ensure continuous improvement, compliance, and responsible data use.



**THANK YOU
FOR YOUR
ATTENTION**

Questions and feedback welcome